ASSIGNMENT COVERSHEET

UTS

UTS: ENGINEERING & INFORMATION TECHNOLOGY

SUBJECT NUMBER & NAME	NAME OF STUDENT(s) (PRINT CLEARLY)		STUDENT ID(s)
32130 Fundamentals of Data Analytics Autumn 2025	*Srichairattanakul* *SURNAME*	*Warit* *FIRST NAME*	25745973

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NAME OF TUTOR	TUTORIAL GROUP	DUE DATE
		04/04/2025

ASSESSMENT ITEM NUMBER & TITLE

Assessment Task 2: Data exploration and preparation

I acknowledge that if AI or another nonrecoverable source was used to generate materials for background research and self-study in producing this assignment, I have checked and verified the accuracy and integrity of the information used.
I confirm that I have read, understood and followed the guidelines for assignment submission and presentation on page 2 of this cover sheet.
I confirm that I have read, understood and followed the advice in the Subject Outline about assessment requirements.
I understand that if this assignment is submitted after the due date it may incur a penalty for lateness unless I have previously had an extension of time approved and have attached the written confirmation of this extension.

Declaration of originality: The work contained in this assignment, other than that specifically attributed to another source, is that of the author(s) and has not been previously submitted for assessment. I have rewritten any material provided by AI or other nonrecoverable sources and where appropriate acknowledged their contribution. I understand that, should this declaration be found to be false, disciplinary action could be taken and penalties imposed in accordance with University policy and rules. In the statement below, I have indicated the extent to which I have collaborated with others, whom I have named.

No content generated by AI technologies or other sources has been presented as my own work and I have rewritten any text provided by AI or other sources in my own words.

Statement of collaboration:

Signature of student(s) ____Warit Srichairattanakul_____ Date _____3 April 2025_____

ASSIGNMENT RECEIPT

To be completed by the student if a receipt is required

SUBJECT NUMBER & NAME	NAME OF TUTOR

SIGNATURE OF TUTOR	RECEIVED DATE

1A. Initial data exploration

1. Identify the attribute type of each attribute in the dataset.

Attribute	Type	Justification
id	Nominal	ID is a unique identifier for each patient. It is a nominal type because it has no numerical meaning for identification, and basic statistical calculations such as mean, median, and standard deviation are meaningless.
age	Ratio	Age is measured in real numbers, ratio calculations are meaningful and can be calculated for the average or difference in age, so it is considered a ratio attribute.
gender	Nominal	The values are just labels that categorise data by their name or explanation. They do not have a numerical meaning.
height	Ratio	Height is measured in real numbers, and ratio calculations are meaningful, for example, A is 170 cm tall and B is 160 cm tall. A is 1.0625 times taller than B, which is meaningful in ratio and height has a clear definition of zero. A value of 0 cm means "no height" or physical absence.
weight	Ratio	Weight is measured in real numbers, and ratio calculations are meaningful, for example, A weighs 50 kg and B weighs 25 kg. A is 2 times heavier than B, which is meaningful in ratio and weight has a clear definition of zero. The value 0 kg means "weightless" or no mass.
ap_hi	Ratio	It is measured in mmHg, has a true zero (0 mmHg means no pressure), and has meaningful comparisons, such as a person with blood pressure 180 mmHg is 2 times higher than a person with blood pressure 90 mmHg.
ap_lo	Ratio	It is measured in mmHg, has a true zero (0 mmHg means no pressure), and has meaningful comparisons.
cholesterol	Ordinal	The data is placed in a specific order. While each order is ranked from Well-above normal > above normal > normal, there is no information that there are differences between each value. The addition and subtraction do not make sense.
gluc	Ordinal	
smoke	Nominal	The values are just labels that categorise data by their name, explanation, or numerical value. They do not have a numerical meaning. For example, there is no meaning in switching the order between smoke and non-smoke.
alco	Nominal	
active	Nominal	
cardio	Nominal	

2. Identify the values of the summarising properties for the attributes, including frequency, location, and spread

This is the statistics summary for each attribute on this dataset.

Name	Mini...	Maxi...	25% Q...	50% ...	75% ...	Mean	Mean A...	Standar...	Sum	Variance	Skewness
age	10,964	23,692	17,649.5	19,701	21,316	19,456.362	2,056.828	2,465.391	217,911,251	6,078,154.596	-0.319
gender	1	2	1	1	2	1.342	0.45	0.474	15,032	0.225	0.665
height	64	198	159	165	170	164.347	6.233	8.14	1,840,689	66.253	-0.821
weight	23	178	65	72	82	74.186	11.02	14.277	830,880.7	203.825	0.914
ap_hi	10	1,400	120	120	140	127.225	14.06	29.375	1,424,918	862.911	22.417
ap_lo	0	9,011	80	80	90	98.667	34.888	203.814	1,105,073	41,540.302	28.408
Age_year	30.038	64.91	48.355	53.975	58.4	53.305	5.635	6.754	597,017.126	45.623	-0.319

Figure 1: Table of statistics for each attribute

This dataset provides statistical summaries of attributes from a sample of 11,200 people. Key insights include:

1. ID

The values are the identification number of the sample. These numbers are only assigned when a new sample record is added. There is no meaning to find any statistical calculation from this value because the number of ID:1000 does not mean it is higher or lower than the number of ID:2000. Subtraction of the highest ID and the lowest ID does not mean this result is the total number of ID because the value did not appear consistently.

id ↑ Number (integer)	id ↓ Number (integer)
13	99998
25	99975
45	99974
61	99973
77	99972
83	99964
92	99960
114	99959
124	99948
126	99940
131	99932
136	99930
142	99923

Figure 2: Sorting of IDs

2. Age

The sample's age range is 34.8872 years, between 30.038 and 64.91 years (10,964 and 23,692 days). The 25th percentile (Q1) value is 48.355 years, and the 75th percentile (Q3) value is 58.4 years, which means 50 percent of the sample lies between 48.355 and 58.4 years old. The mean is 53.305 years and the median is 53.975 years. The variance and standard deviation are 45.623 and 6.754, respectively, which means that the ages of the sample are moderately spread around the mean.

The graph is left-skewed and the skewness value is -0.319, which means the population on the dataset consists of older people more than young people, but not much.

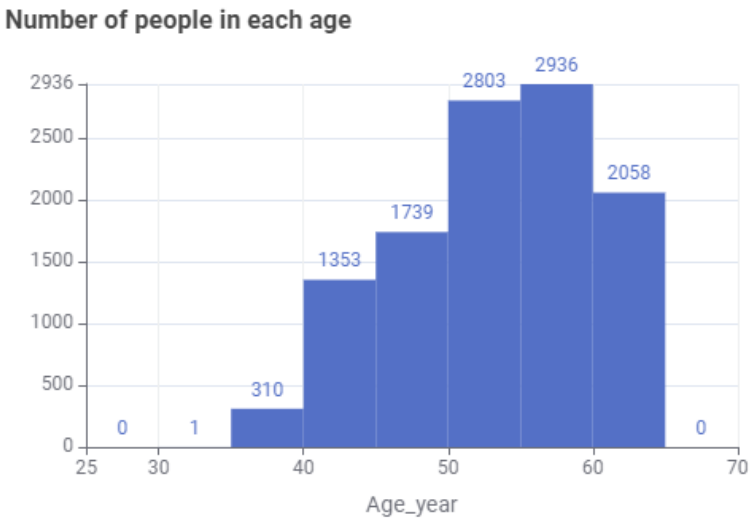

Figure 3: Histogram showing the distribution of age

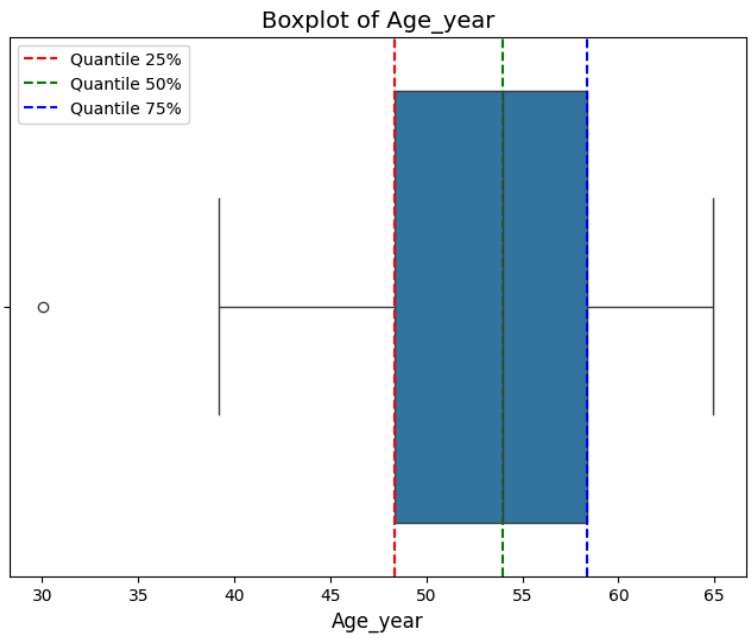

Figure 4: Boxplot showing the distribution of age

3. Gender

The sample of this dataset consists of 2 genders, male and female. The number of female and male samples in the dataset is 7,368 persons and 3,832 persons, respectively. The ratio of females to males in the dataset is approximately 65.8% females and 34.2% males.

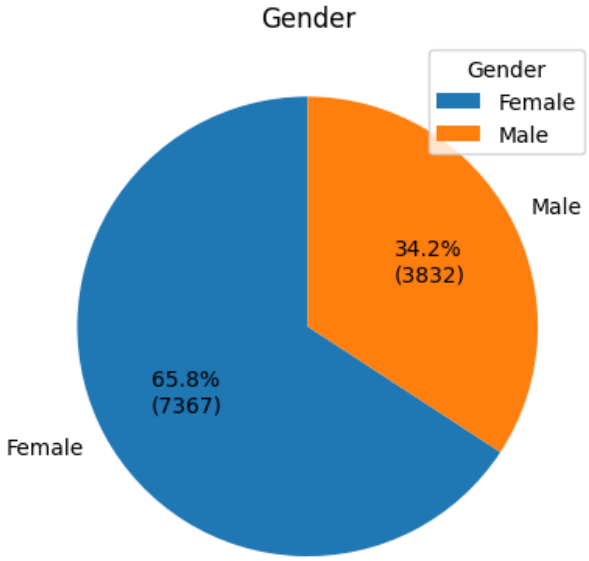

Figure 5: Pie chart showing the number of each gender

4. Height

The sample's height range is 134 centimeters, between 64 and 198 centimeters. The 25th percentile (Q1) value is 159 centimeters, and the 75th percentile (Q3) value is 170 centimeters, which means 50 percent of the sample's height is between 159 and 170 centimeters. The mean is 164.347 centimeters and the median is 165 centimeters. The variance and standard deviation are 66.253 and 8.14, respectively, which means that the ages of the sample are moderately spread around the mean.

The graph is symmetric, and the skewness value is -0.821, which means the population on the dataset consists of taller people than shorter people, but not much.

The number of outliers when using the formula to find outliers is too much. When I think about it realistically, the height of normal people can be around 140 to 190 centimeters, so I changed the factor from 1.5 to 2.5 to find just the extreme outliers from the dataset.

From: UpperBound = Q3 + 1.5*IQR and LowerBound = Q1 - 1.5*IQR

To: UpperBound = Q3 + 2.5*IQR and LowerBound = Q1 - 2.5*IQR

From the boxplot graph, since this dataset contains a lot of sample data of people with a height of less than 140 cm, I think that this dataset may be biased by using data from the sample group that may have dwarfism in this dataset.

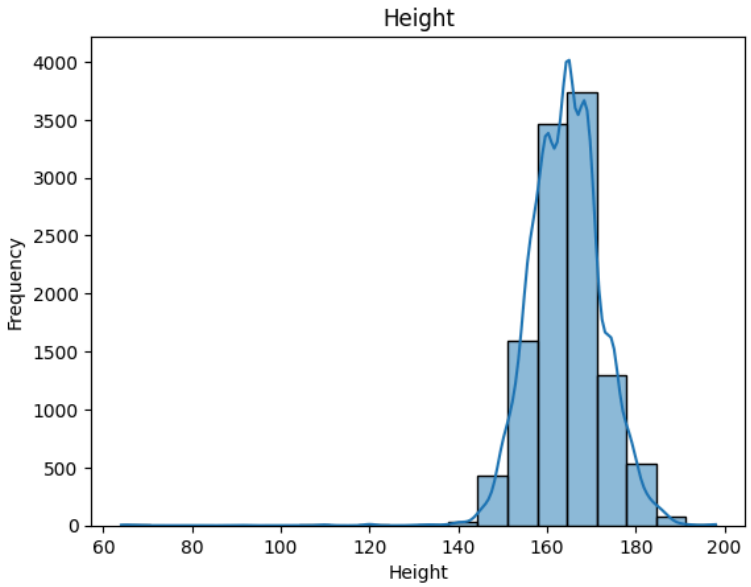

Figure 6: Histogram showing the distribution of height

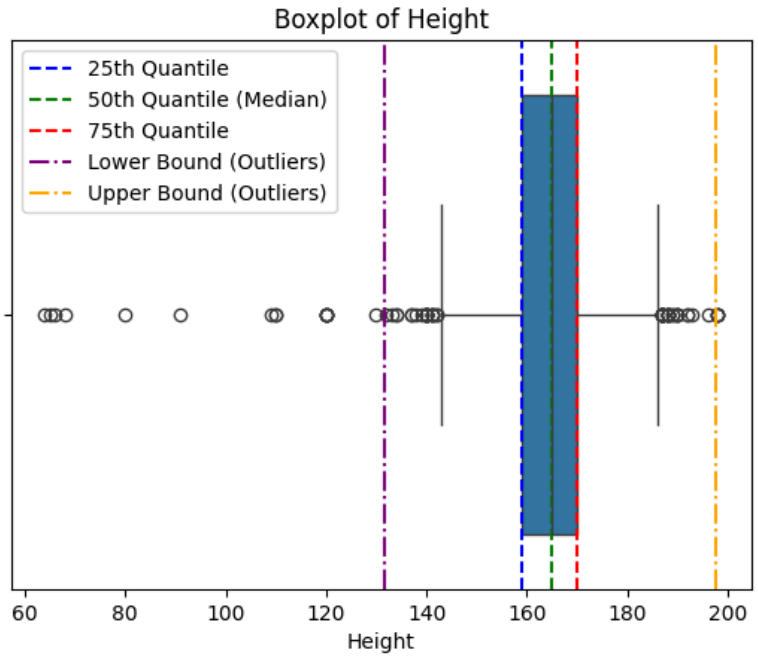

Figure 7: Boxplot showing the distribution of height

5. Weight

The sample's weight range is 155 kg, between 23 and 178 kg. The 25th percentile (Q1) value is 65 kg, and the 75th percentile (Q3) value is 82 kg, which means 50 percent of the sample's weight lies between 65 and 82 kg. The mean is 74.186 kg and the median is 72 kg. The spread of weights around the mean is moderate, as indicated by the standard deviation at 14.277 and the variance at 203.825.

The skewness value of 0.914 indicates that the distribution of the sample's weights is slightly right-skewed.

The number of outliers when using the normal formula to find outliers is too much but considering reality, I think the weight of normal people should be around 40 to 100, so I have not changed the formula and consider that this dataset may be biased by using data from the sample group that may have obesity.

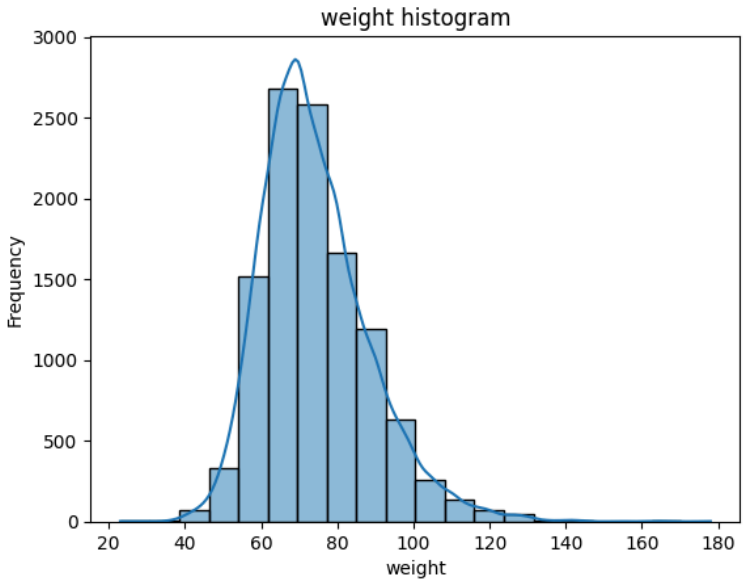

Figure 8: Histogram showing the distribution of weight

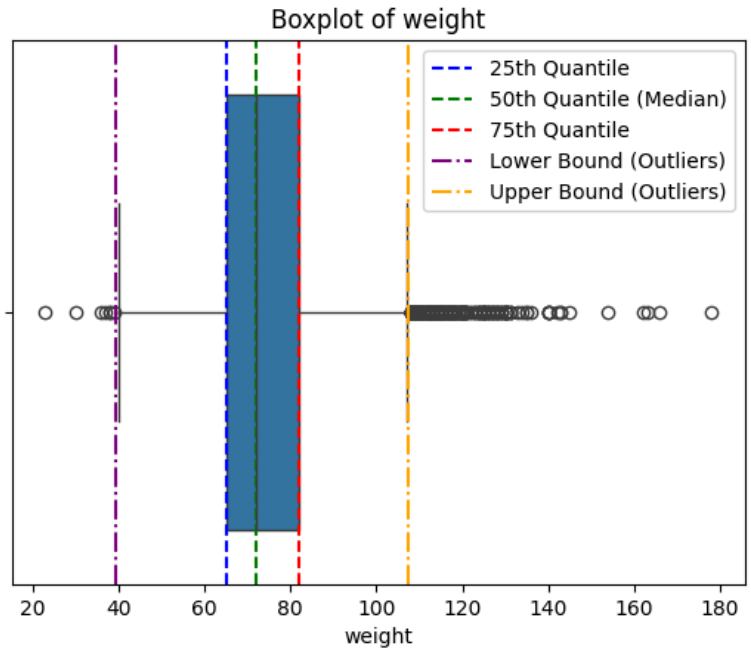

Figure 9: Boxplot showing the distribution of weight

6. ap_hi

The sample's systolic blood pressure (ap_hi) range is 1390 mmHg, between 10 and 1400 mmHg. The 25th percentile (Q1) value is 120 mmHg, and the 75th percentile (Q3) value is 140 mmHg, which means 50 percent of the sample's weight lies between 120 and 140 mmHg. The mean is 127.225 mmHg and the median is 120 mmHg. The spread of the data around the mean is high, as indicated by the standard deviation of 29.375 and the variance of 862.911.

A skewness value of 22.417 is extremely high and indicates that the distribution of the sample's systolic blood pressures is highly right-skewed, meaning there are extreme outliers with significantly higher systolic blood pressure. Since the data is skewed and influenced by outliers, the median (120 mmHg) is a better measure of central tendency than the mean.

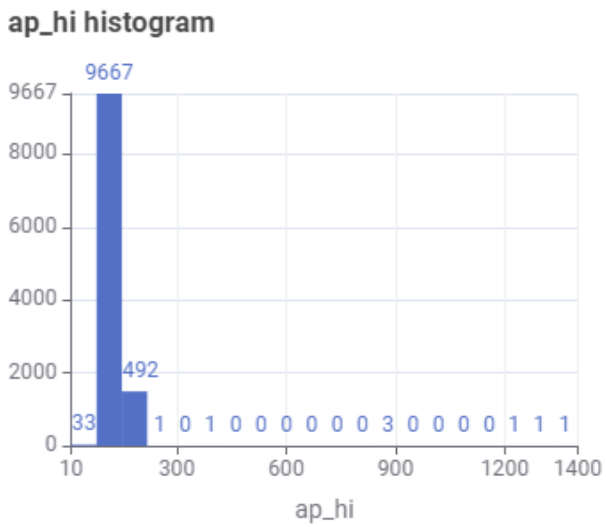

Figure 10: Histogram showing the distribution of systolic blood pressure

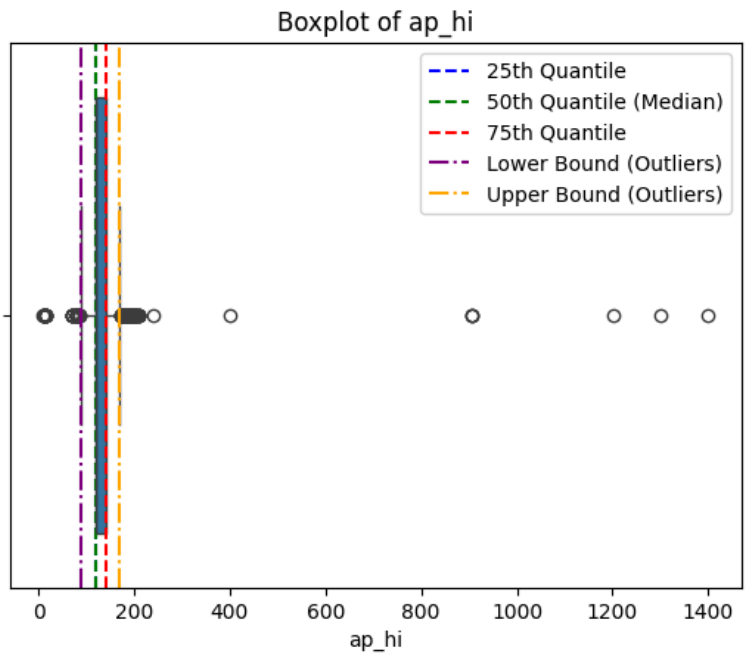

Figure 11: Boxplot showing the distribution of systolic blood pressure

7. ap_lo

The sample's diastolic blood pressure (ap_lo) has a range of 9011 mmHg, with values ranging from 0 to 9011 mmHg. The 25th percentile (Q1) value is 80 mmHg, and the 75th percentile (Q3) value is 90 mmHg, which means that 50% of the sample's diastolic blood pressure values lie between 80 and 90 mmHg. The mean diastolic blood pressure is 98.667 mmHg, and the median is 80 mmHg.The spread of the data around the mean is extremely high, as indicated by the standard deviation of 203.814 and the variance of 41,540.302.

A skewness value of 28.408 is extremely high and indicates that the distribution of the sample's diastolic blood pressure is highly right-skewed, meaning that there are extreme outliers with significantly higher diastolic blood pressure. Since the data is skewed and influenced by outliers, the median (90 mmHg) is a better measure of central tendency than the mean.

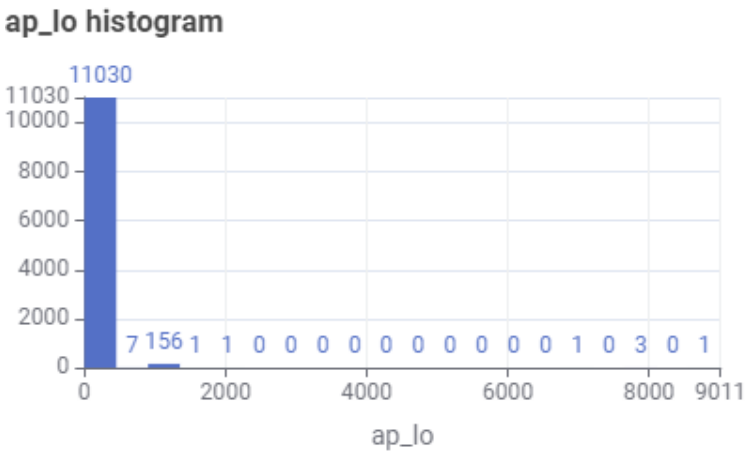

Figure 12: Histogram showing the distribution of diastolic blood pressure

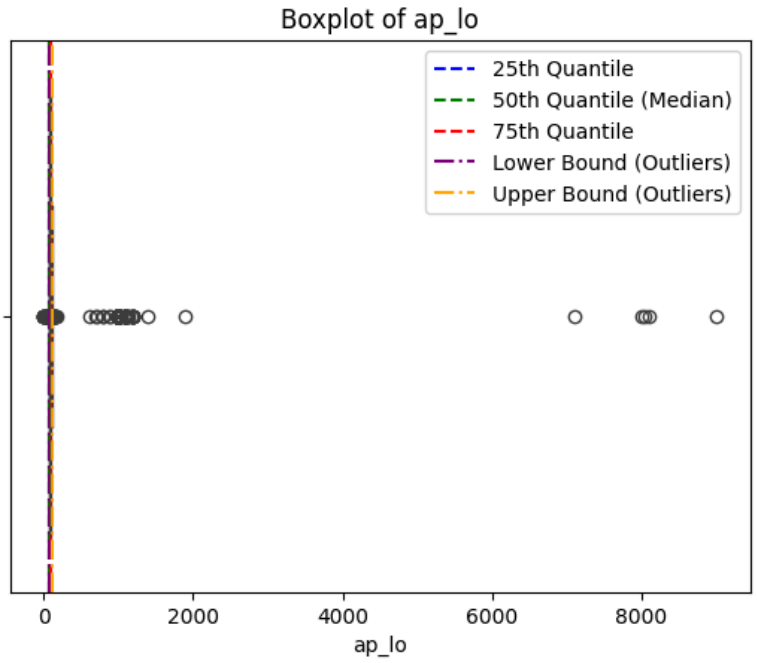

Figure 13: Boxplot showing the distribution of diastolic blood pressure

8. Cholesterol

The sample of this dataset consists of 3 levels of cholesterol: normal, above normal, and well above normal. The number of patients with a normal cholesterol level is 8,368 persons or 74.7%, above normal cholesterol is 1,289 persons or 11.5%, and well above normal cholesterol level is 1,543 persons or 13.8%.

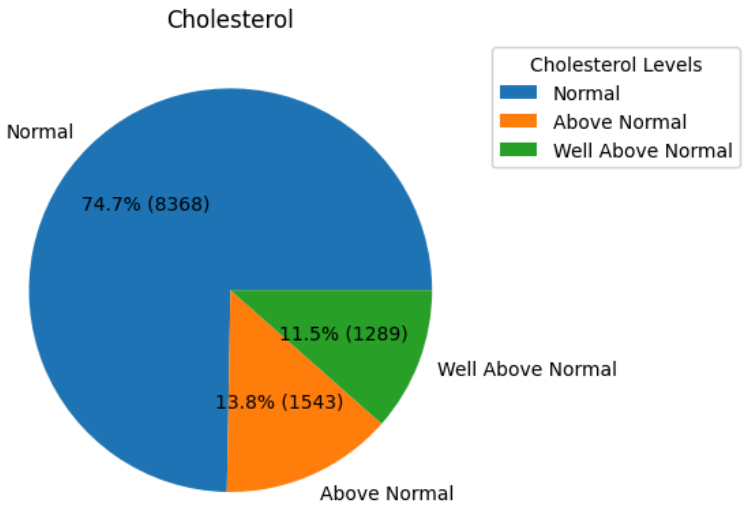

Figure 14: Pie chart showing the number of each cholesterol level

9. Gluc

The sample of this dataset consists of 3 levels of glucose: normal, above normal, and well above normal. The number of patients with a normal glucose level is 9,475 persons or 84.6%, above normal glucose level is 857 persons or 7.7%, and well above normal glucose level is 868 persons or 7.8%.

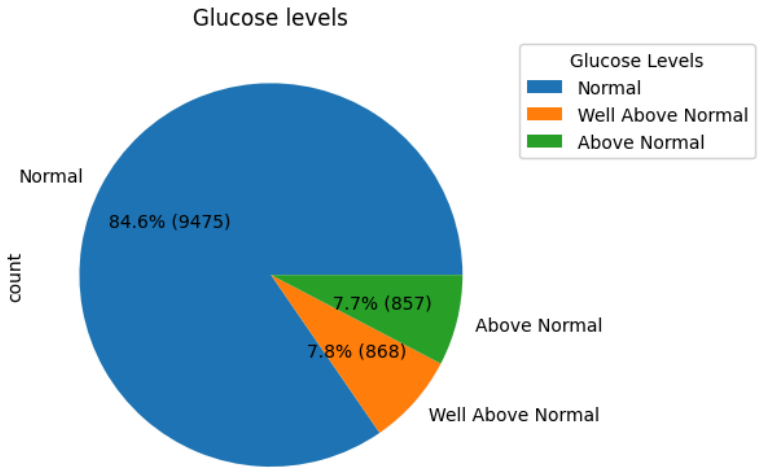

Figure 15: Pie chart showing the number of each glucose level

10. Smoke

The sample in this dataset is divided into 2 categories: those who smoke and not smoke. The number of samples that are smokers is 976 persons or 8.7% of the sample. The number of sample that is not a smoker is 10,224 persons or 91.3% of the sample.

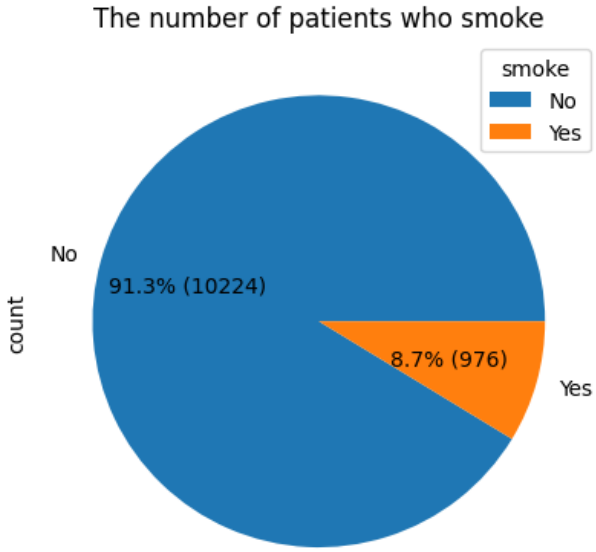

Figure 16: Pie chart showing the number of patients who smoke

11. Alco

The sample in this dataset is divided into 2 categories: those who take alcohol and not take alcohol. The number of samples that are taken alcohol is 10,621 persons, or 94.8% of the sample. The number of samples that do not take alcohol is 579 persons, or 5.2% of the sample.

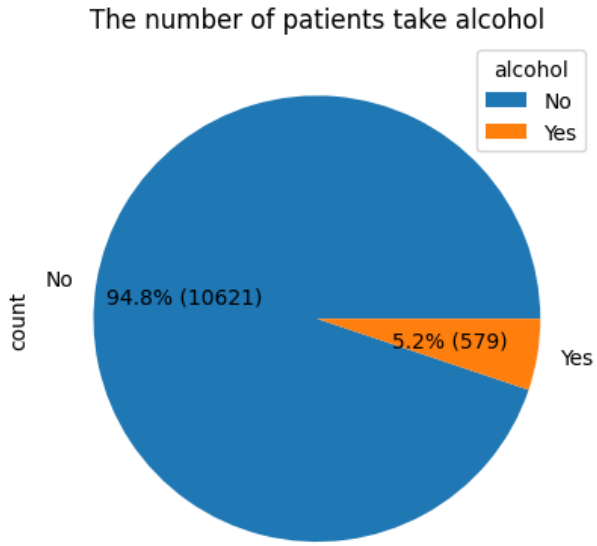

Figure 17: Pie chart showing the number of patients who take alcohol

12. Active

The sample in this dataset is divided into 2 categories: those who do physical activity and do not. The number of samples that do physical activity is 8,972 persons or 80.1% of the sample. The number of samples that don't do physical activity is 2,228 persons or 19.9% of the sample.

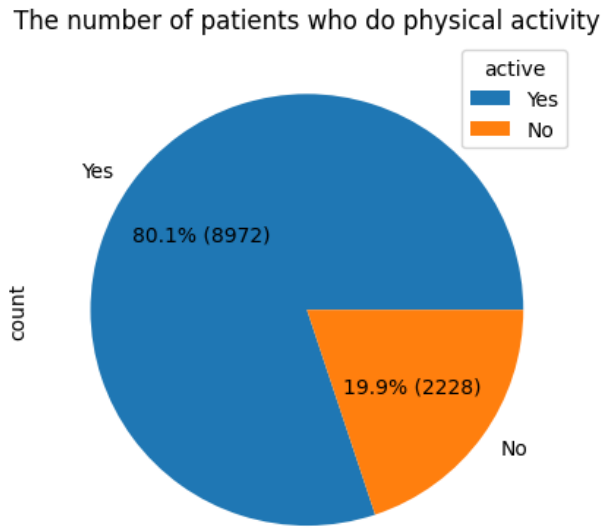

Figure 18: Pie chart showing the number of patients who do physical activity

13. Cardio

The sample in this dataset is divided into 2 categories: those who have cardiovascular disease and don't have cardiovascular disease. The number of samples that have cardiovascular disease is 5,597 persons, or 50% of the sample. The number of samples that don't have cardiovascular disease is 5,603 persons, or 50% of the sample.

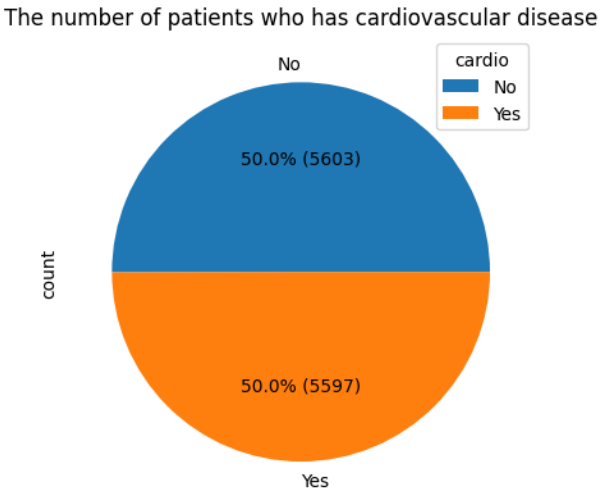

Figure 19: Pie chart showing the number of patients who have cardiovascular disease

3. Explore the relationship between the multiple attributes of the dataset and identify any outliers, clusters of similar instances, "interesting" attributes, and specific values of those attributes.

As shown in Figure 20: Boxplot of the outliers, we can see some outliers from each attribute. Some attributes (e.g., ap_hi and ap_lo) seem to have extreme values that deviate significantly from the main data distribution. These may be measurement errors or genuine outliers. Some data points in height and weight significantly deviate from the real value, such as height more than 200 centimeters and weight lower than 30 kilograms. So I decided to remove these outlier rows before exploring the relationship between attributes.

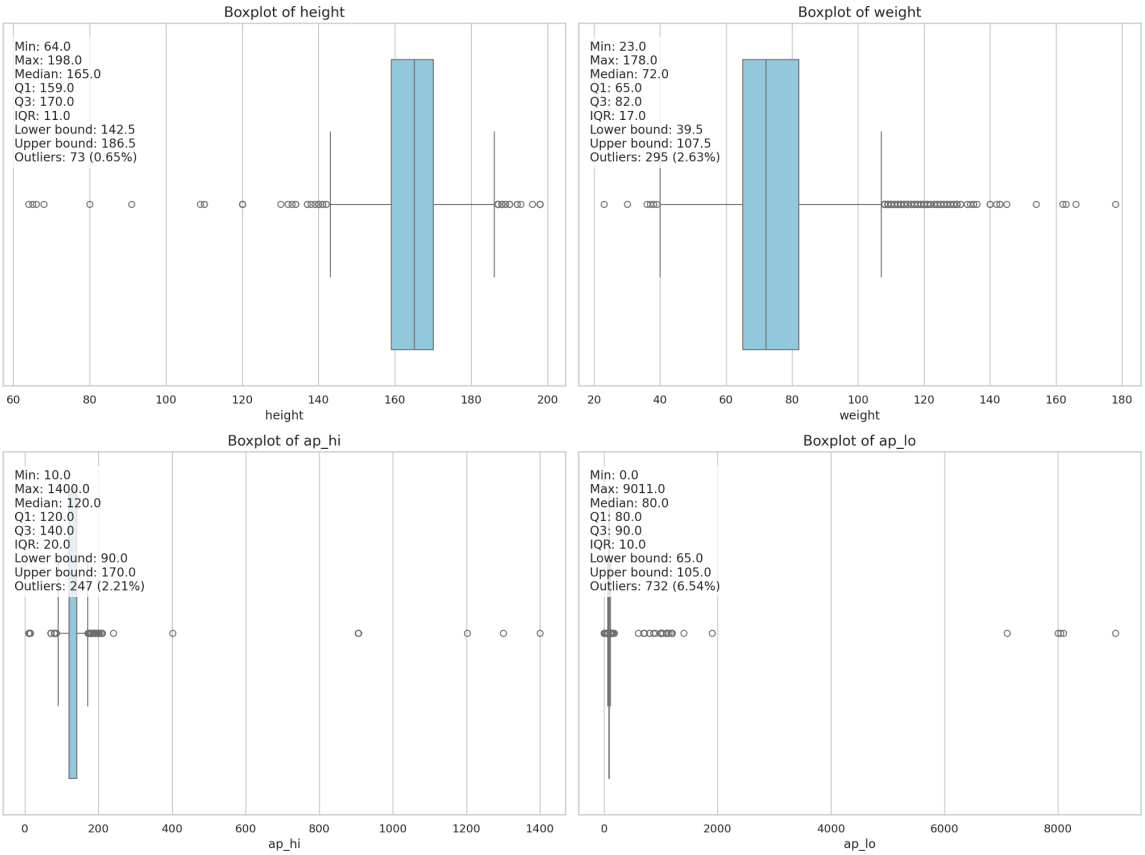

Figure 20: Boxplot of the outliers

To get the attribute relationships, I use the data visualization techniques such as Pairplot to view the scatter plots of every relevant attribute and Correlation Matrix from Heatmap to view the relationship between multiple variables simultaneously.

Because some attributes are categorical(smoke, alco, active, cardio), I must convert them into numeric values (Yes = 1 and No = 0). Then plot the scatter plots for every relevant attribute and the Heatmap correlation matrix.

As shown in Figure 21, Pairplot of every relevant attribute, we can see the relationships between multiple attributes in a dataset. There are some interesting attribute relationships such as

Height and Weight: There is a positive correlation between height and weight. We can see that taller people weigh more than shorter people.

Blood Pressure Values (ap_hi and ap_lo): The systolic pressure (ap_hi) should always be higher than the diastolic pressure (ap_lo). If the data contains instances where ap_lo is greater, there might be data entry errors.

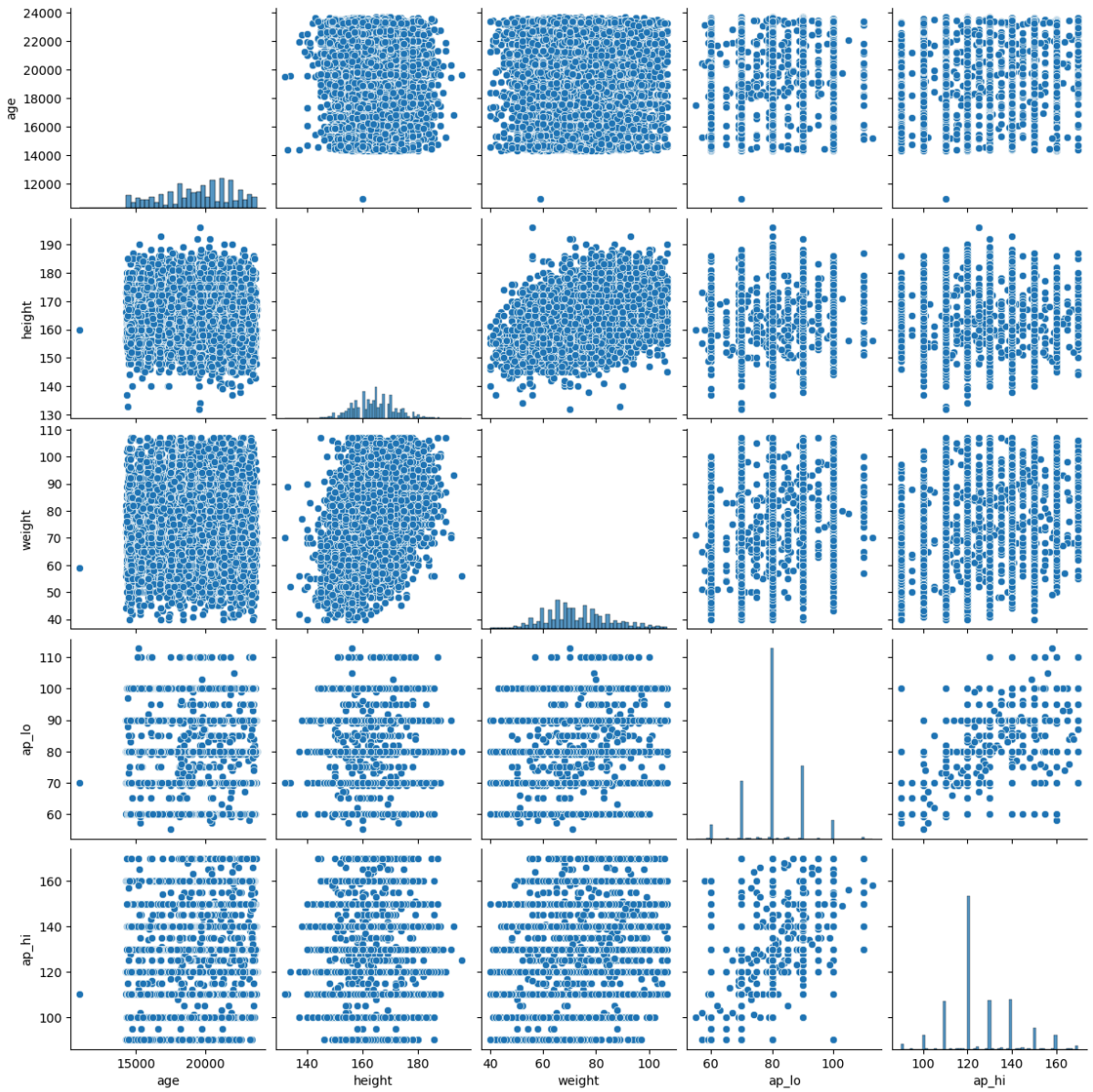

Figure 21: Pairplot of every relevant attribute

As shown in Figure 22: The correlation heatmap, this heatmap was conducted using the Pearson correlation coefficient. We can analyze the relationship between two attributes by looking at the correlation value on the graph. For example, 1.0 means a Perfect positive correlation (as one variable increases, the other increases), -1.0 means a Perfect negative correlation (as one variable increases, the other decreases), and 0.0 means no correlation.

We can see that some attributes are related to each other by observing the graph and the correlation value. Some attributes have a weak correlation value, while some attributes have a strong correlation.

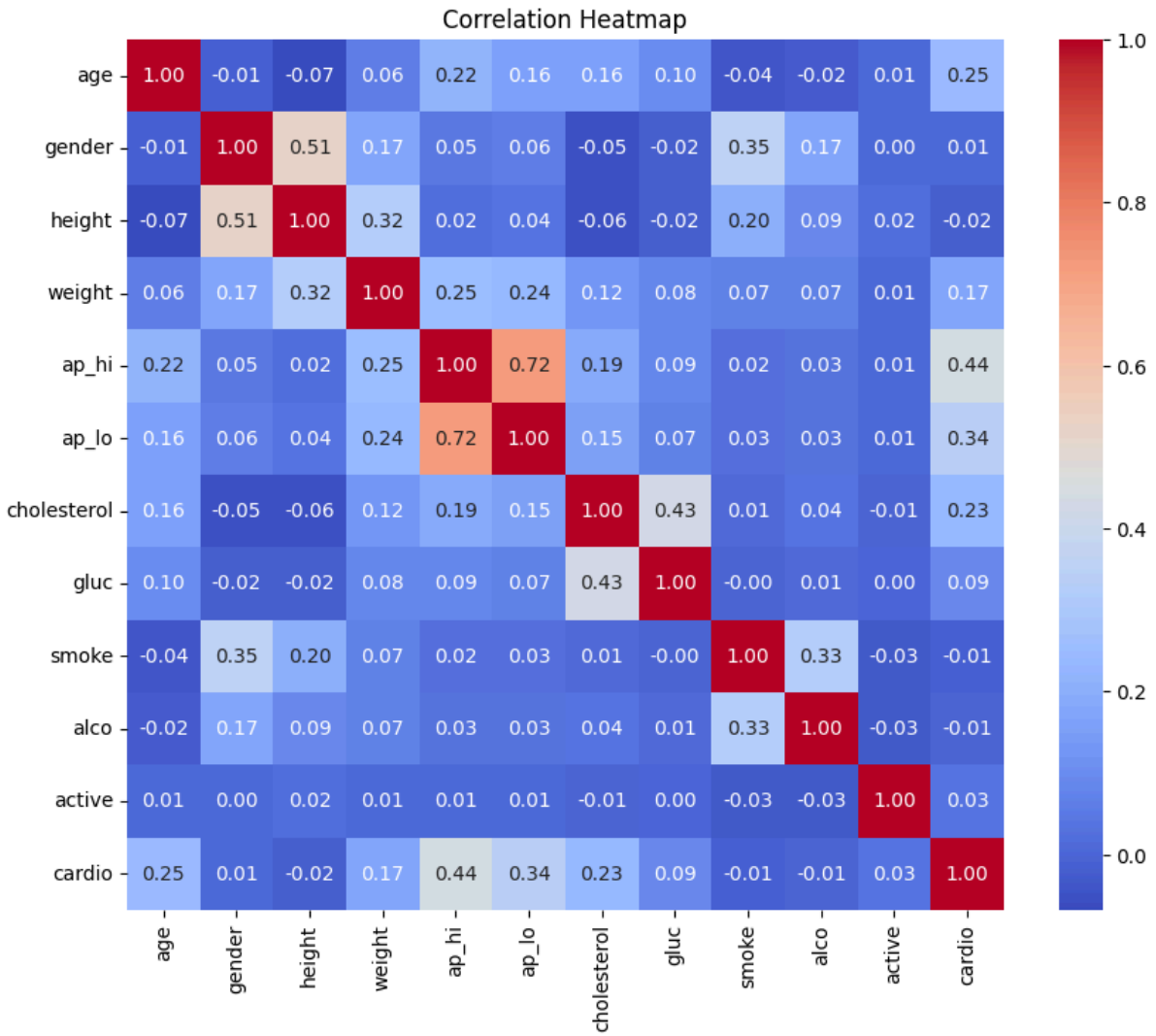

Figure 22: The correlation heatmap

1. Age-related correlation
    - Age and Cardio: The correlation value is 0.25 and is a positive correlation, meaning that the older samples tend to have a higher risk of cardiovascular issues than younger samples.

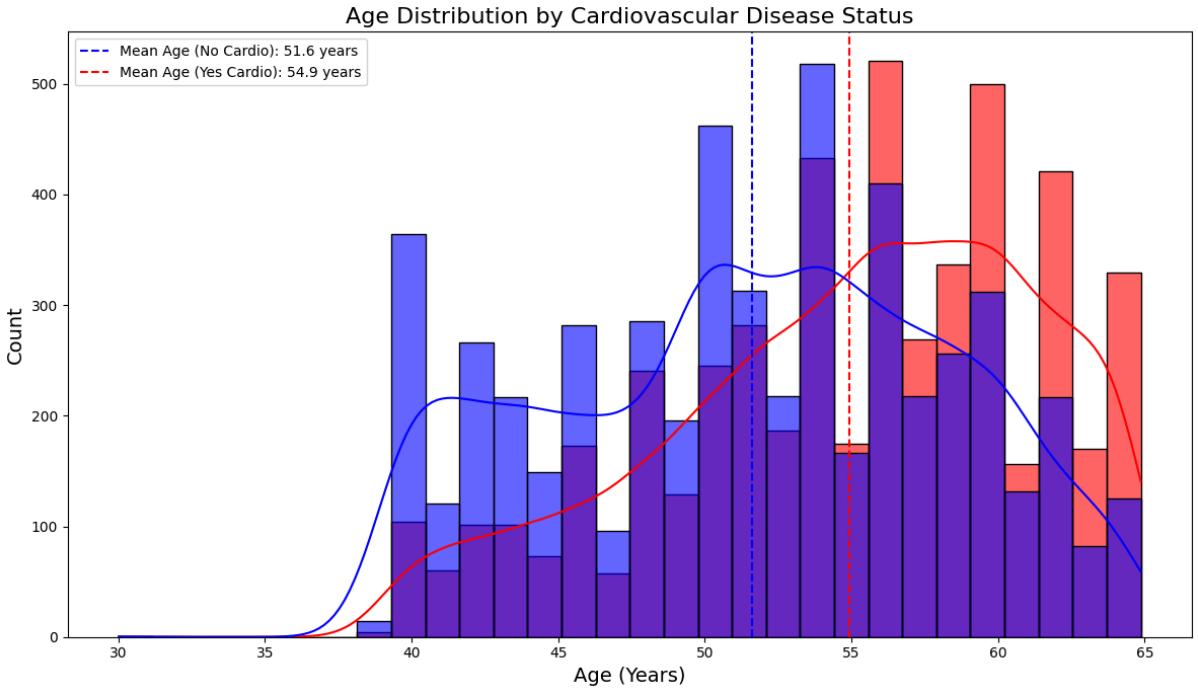

Figure 23: Age Distribution by Cardiovascular Disease

- Age and Gluc: The correlation value is 0.10, meaning that older people tend to have above normal glucose levels.

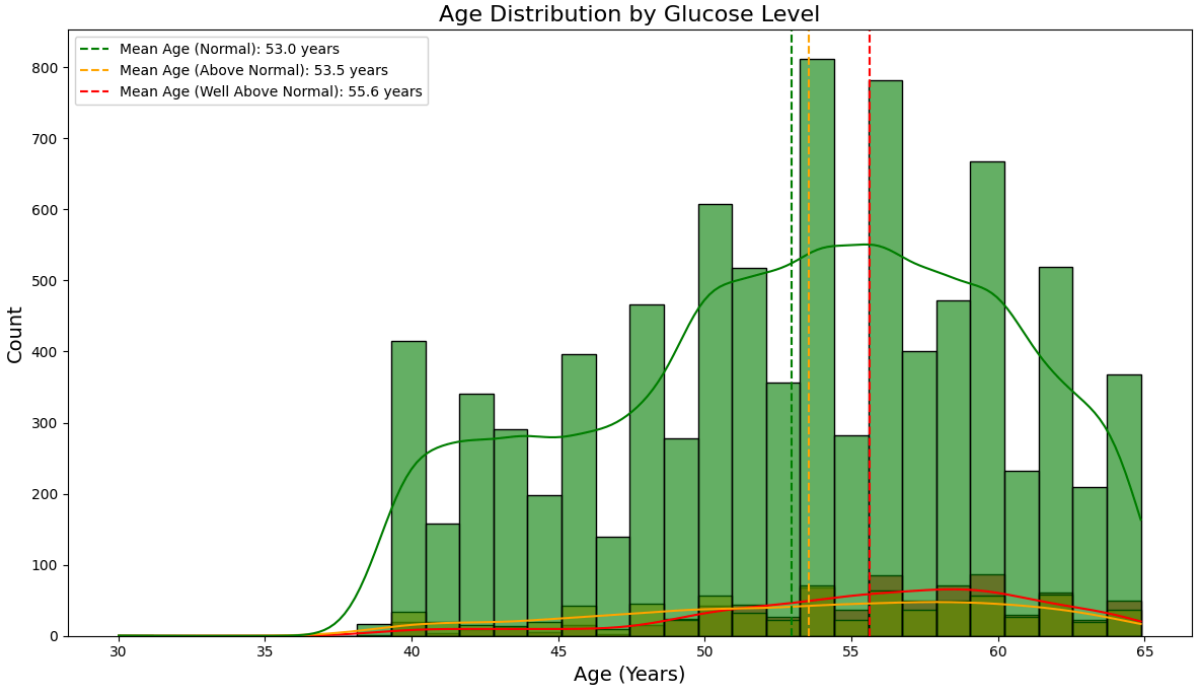

Figure 24: Age Distribution by Glucose Level

- Age and Cholesterol: The correlation value is 0.16 and is a positive correlation, meaning that older people tend to have above normal cholesterol levels.

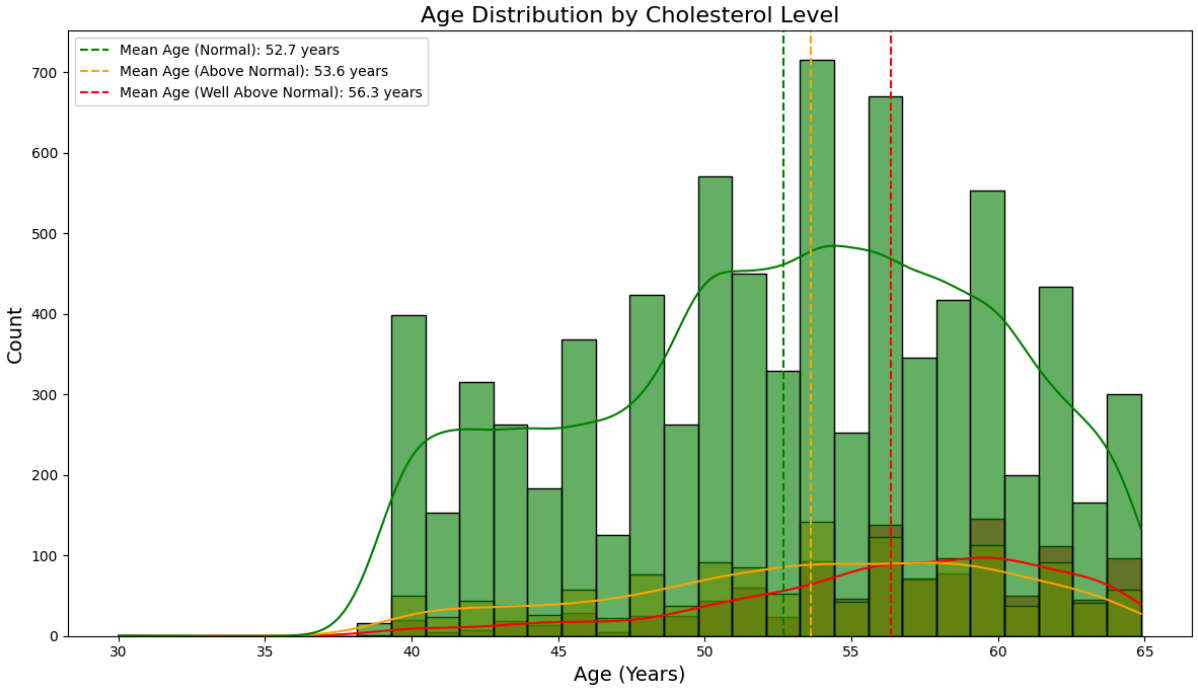

Figure 25: Age Distribution by Cholesterol Level

- Age and Ap_lo: The correlation value is 0.16, meaning that older samples tend to have higher diastolic blood pressure.

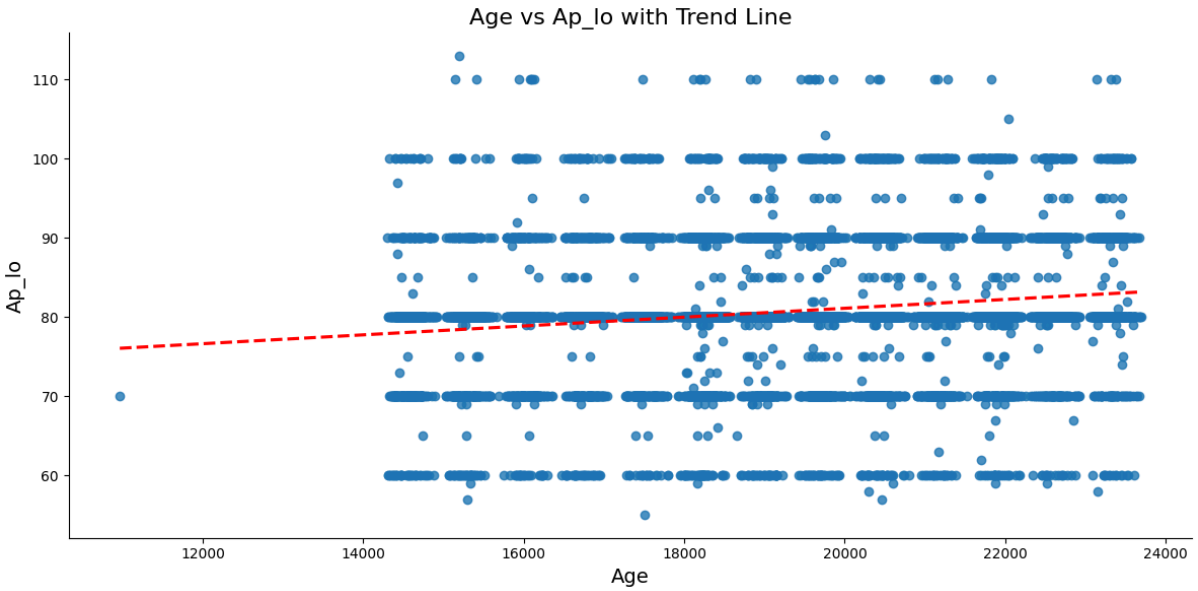

Figure 26: Scatterplot of Age and Ap_lo relationship

- Age and Ap_hi: The correlation value is 0.22, meaning that older samples tend to have higher systolic blood pressure.

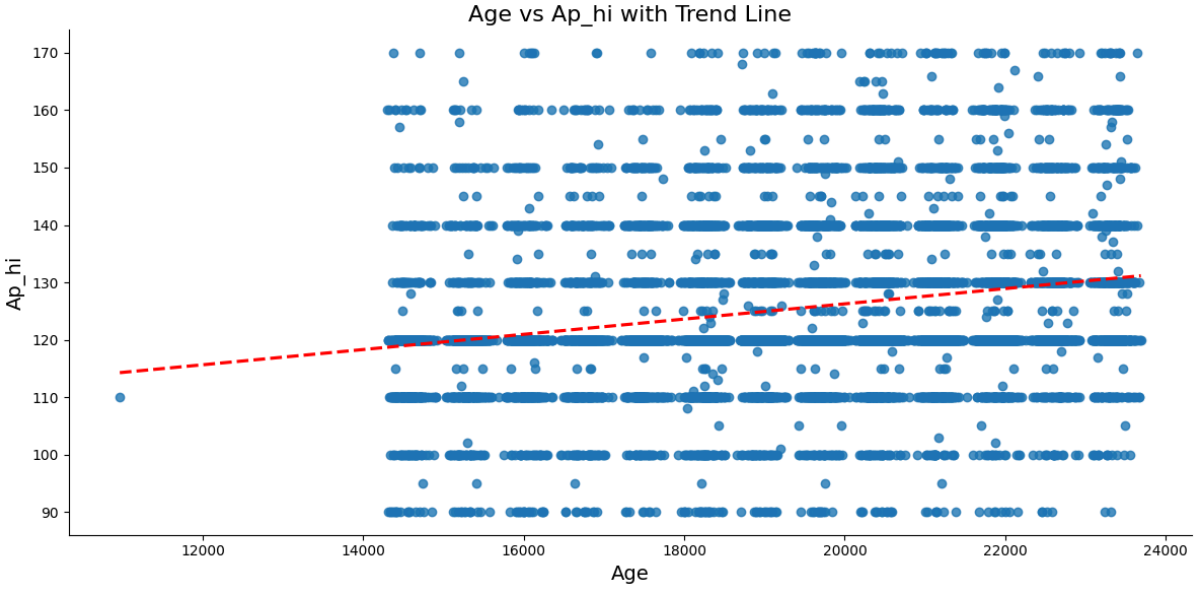

Figure 27: Scatterplot of Age and Ap_hi relationship

2. Gender-relation correlation
    - Gender and Alco: The correlation value is 0.17, meaning that one gender (males) consumes alcohol more frequently.

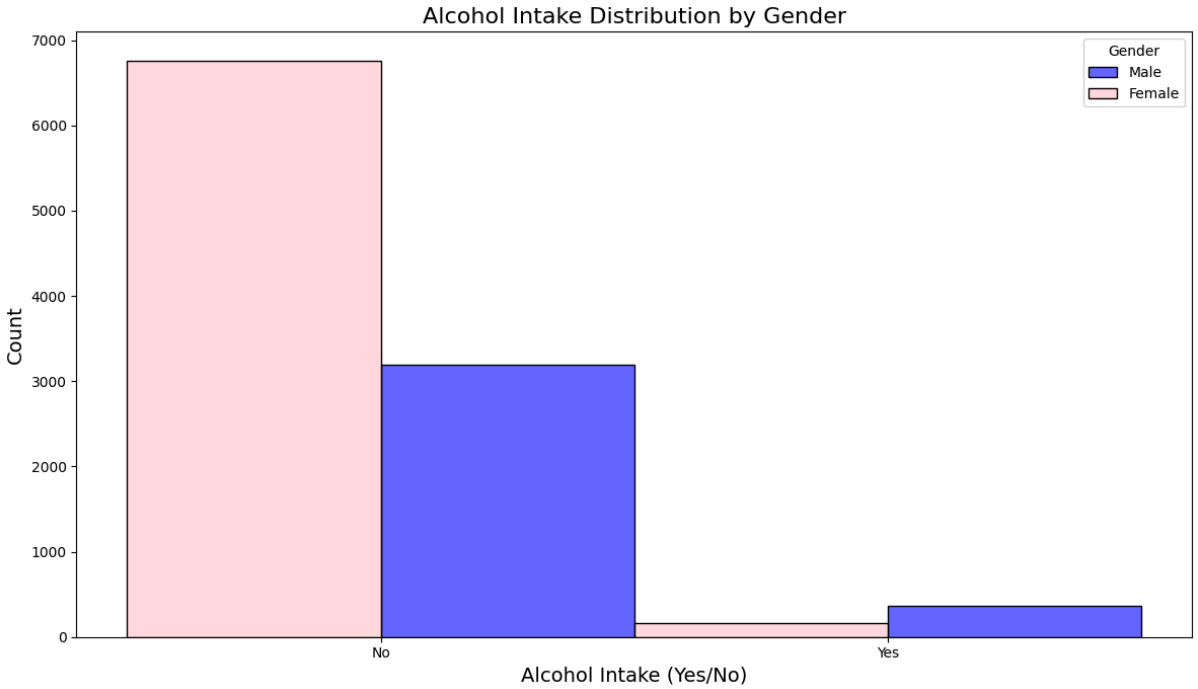

Figure 28: Alcohol Intake distribution by Gender

- Gender and Smoke: The correlation value is 0.35 and is a strong positive correlation, meaning that one gender (males) smokes more frequently than the other.

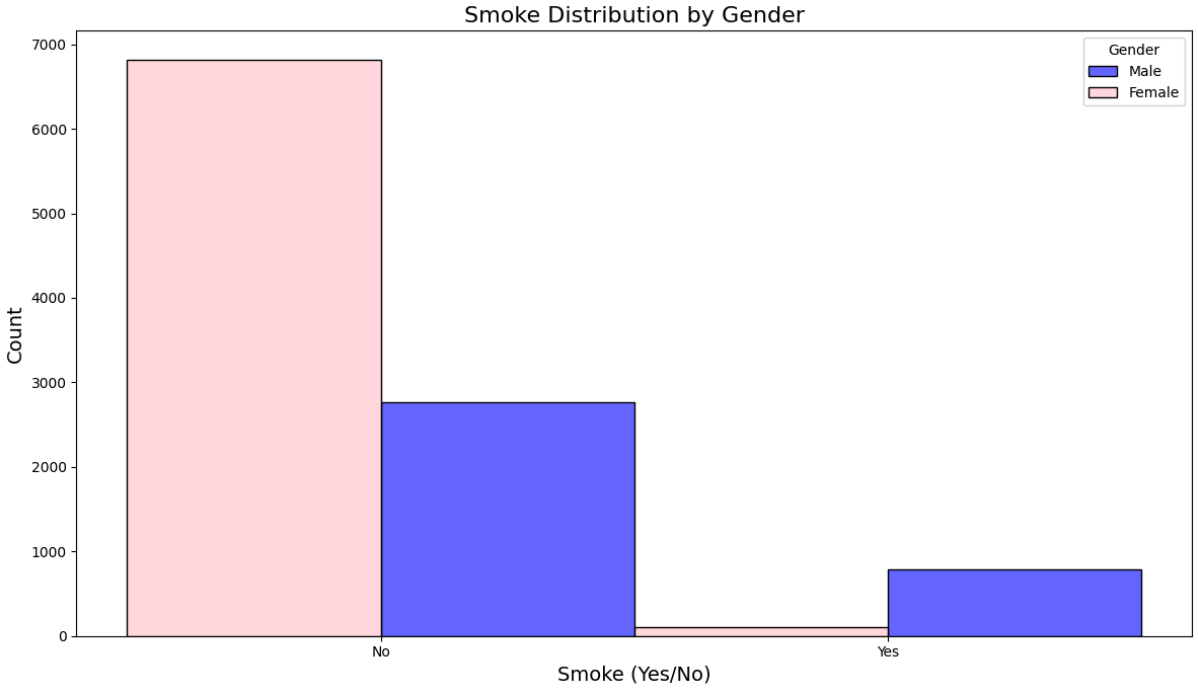

Figure 29: Smoke distribution by Gender

- Gender and Weight: The correlation value is 0.17 has a moderate correlation, possibly due to differences in muscle mass and fat distribution between males and females.

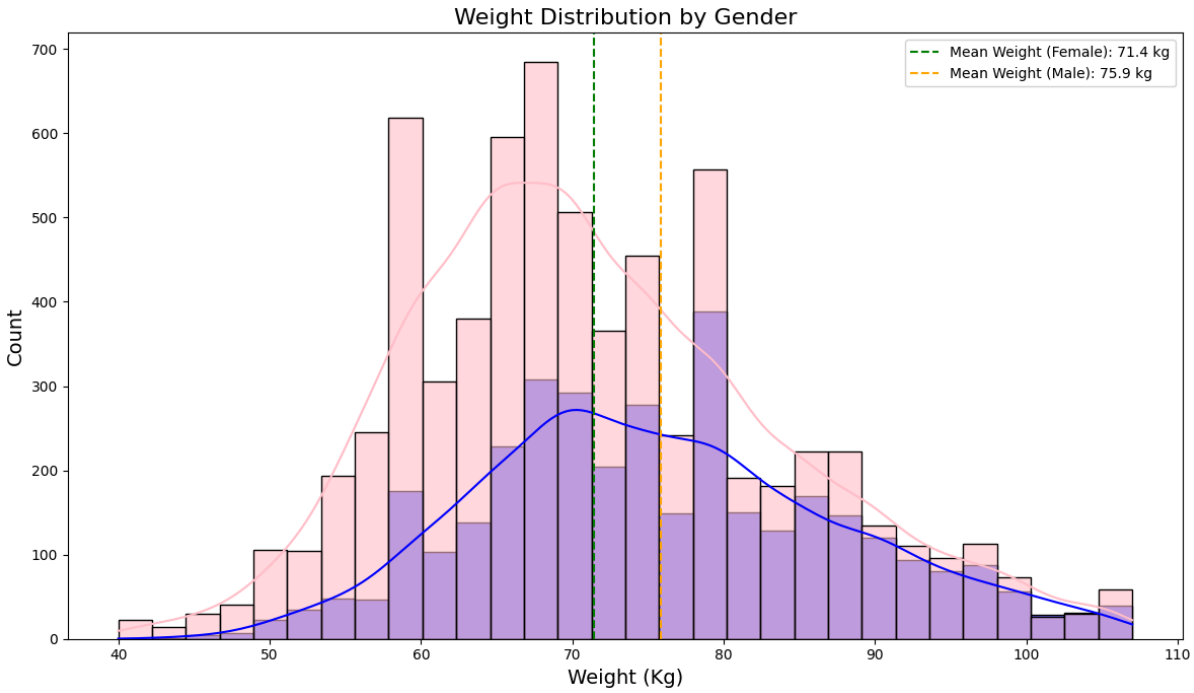

Figure 30: Weight distribution by Gender

- Gender and Height: The correlation value is 0.51 has a strong positive correlation, as men are generally taller than women.

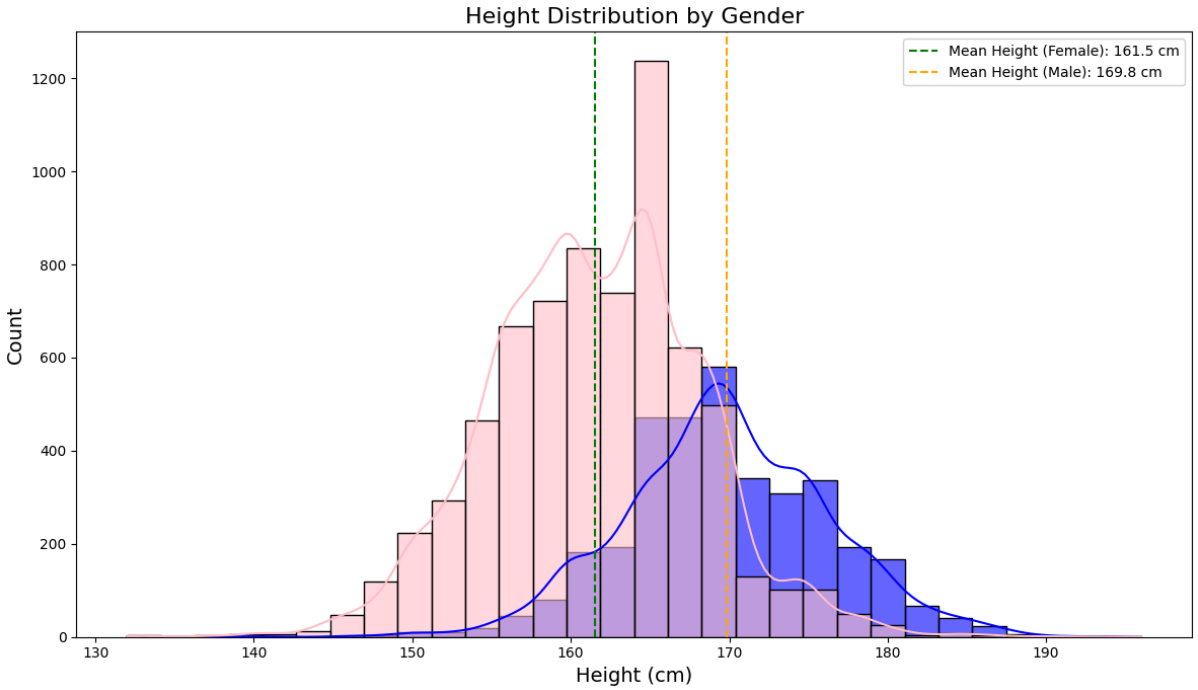

Figure 31: Height distribution by Gender

3. Height-relation correlation
    - Height and Smoke: The correlation value is 0.20 and is a positive correlation. This is an interesting correlation that taller samples smoke more than shorter samples. A possible explanation for this correlation is that men, on average, are taller than women and also tend to smoke more frequently than women, contributing to the observed positive correlation.

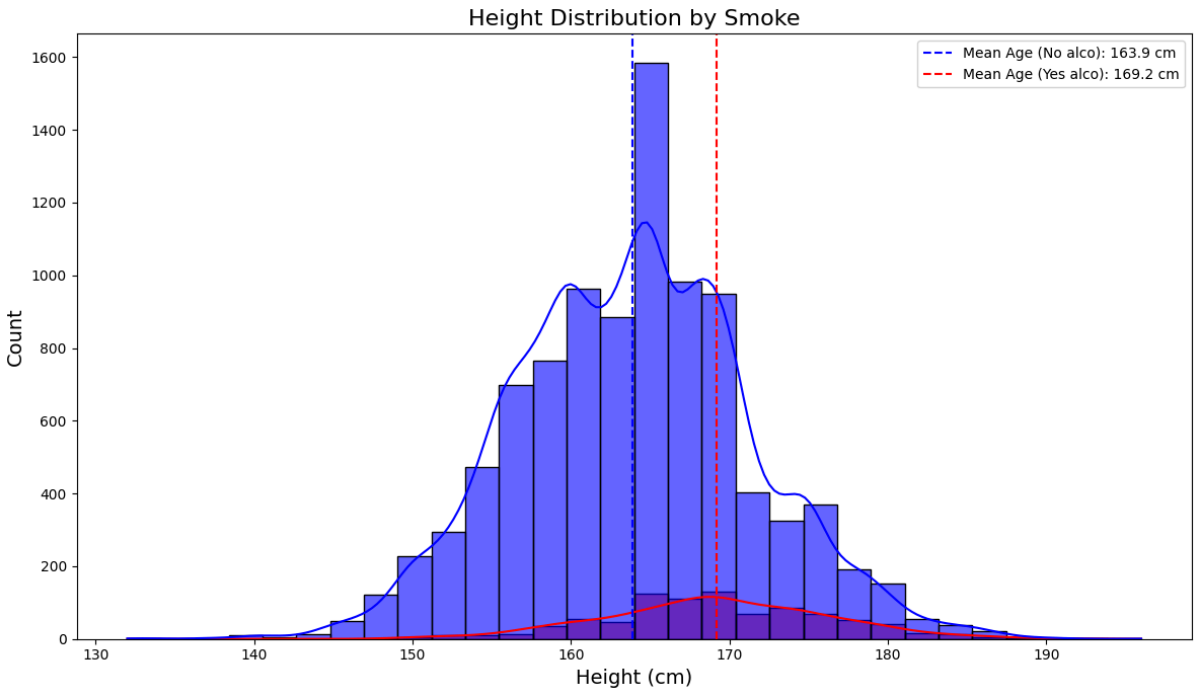

Figure 32: Height distribution by Smoke

- Height and Weight: The correlation value is 0.32 and is a positive correlation, meaning that the taller samples tend to weigh more, which is expected.

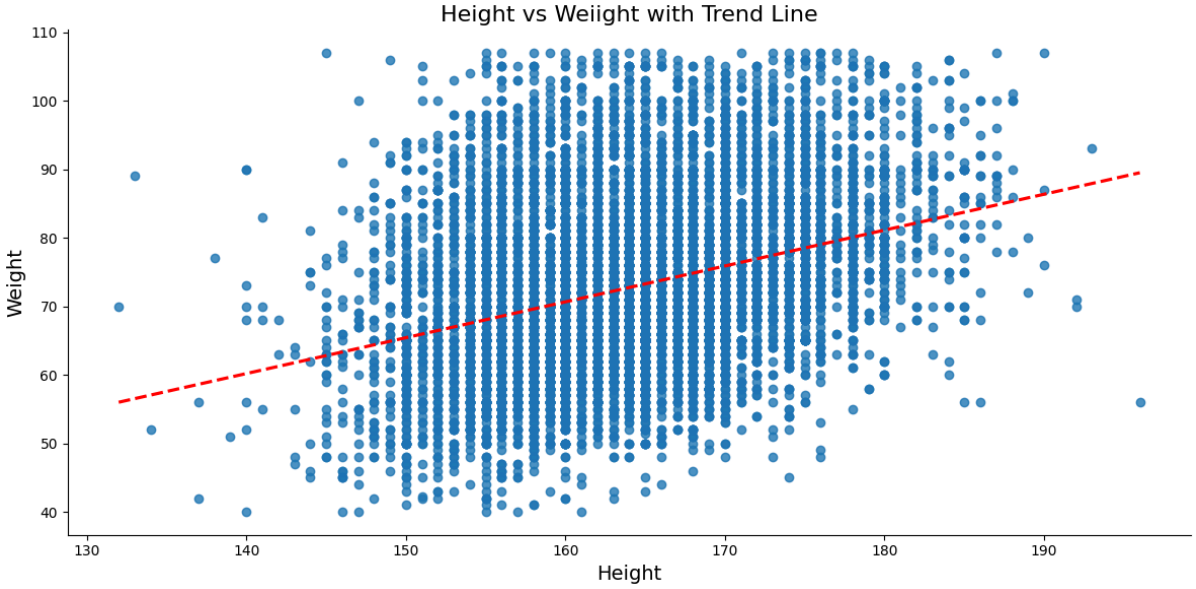

Figure 33: Scatterplot of Height and Weight relationship

4. Weight-relation correlation
    - Weight and Cardio: The correlation value is 0.17, meaning that higher weight is associated with a greater risk of cardiovascular disease.

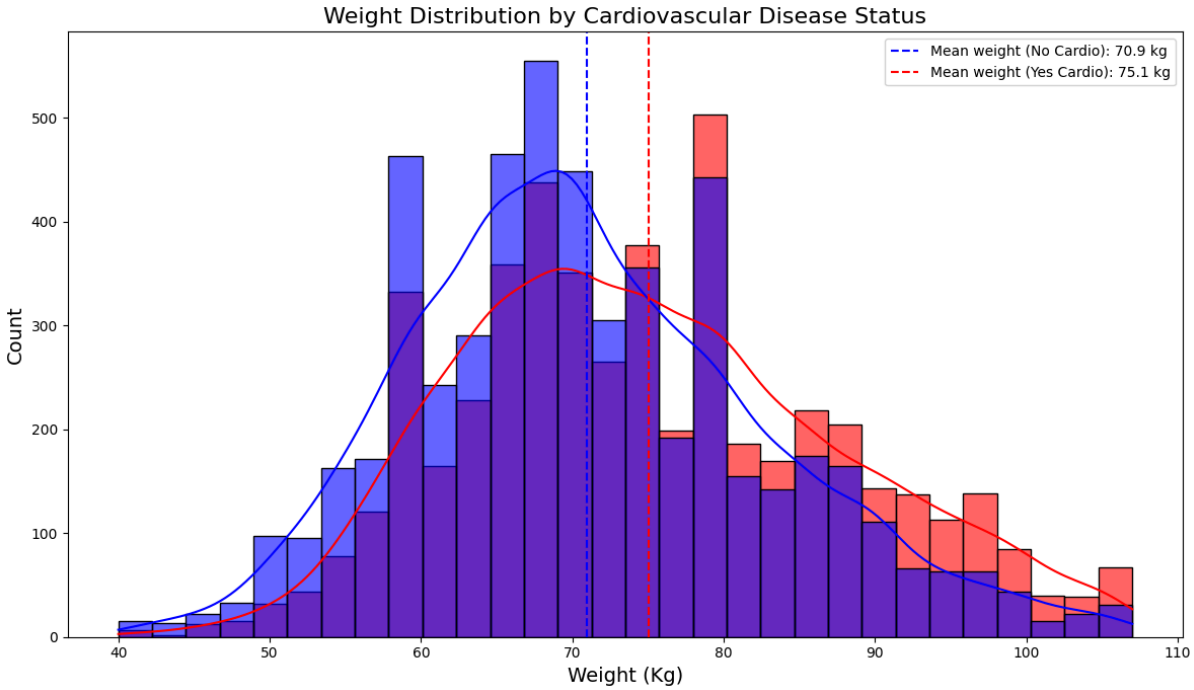

Figure 34: Weight distribution by Cardiovascular disease

- Weight and Cholesterol: The correlation value is 0.12, meaning that people with more weight are likely to have above normal cholesterol levels.

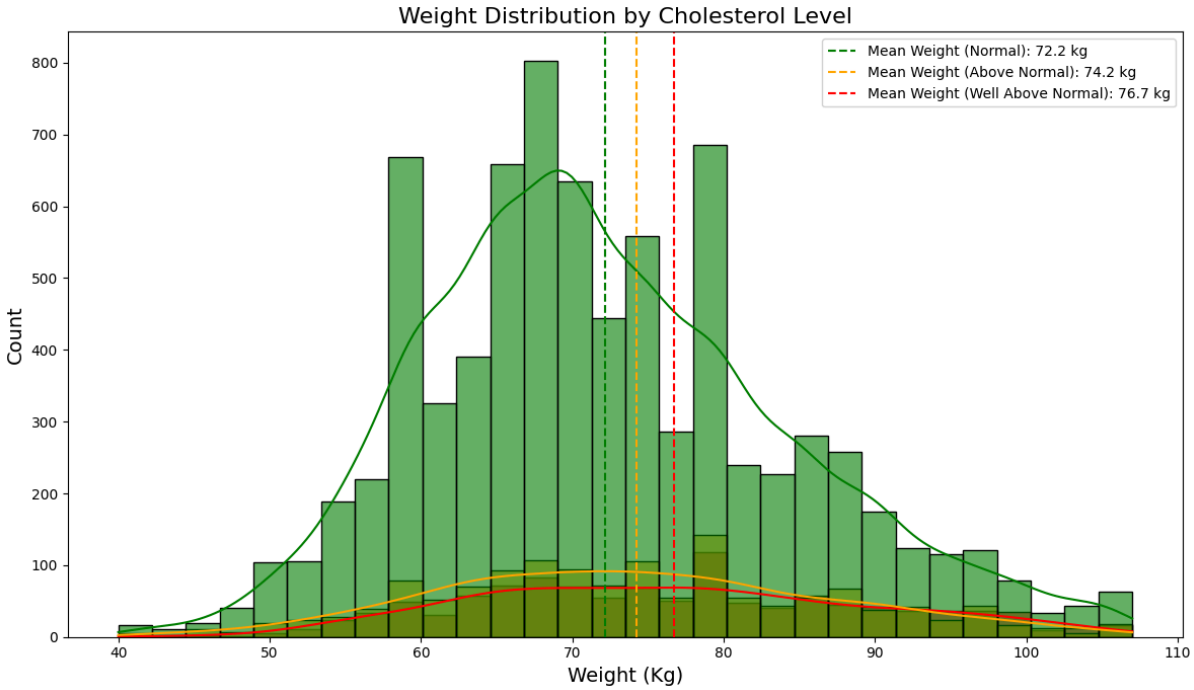

Figure 35: Weight distribution by Cholesterol Level

- Weight and Ap_lo: The correlation value is 0.24, meaning that people who weigh more likely to have higher ap_lo.

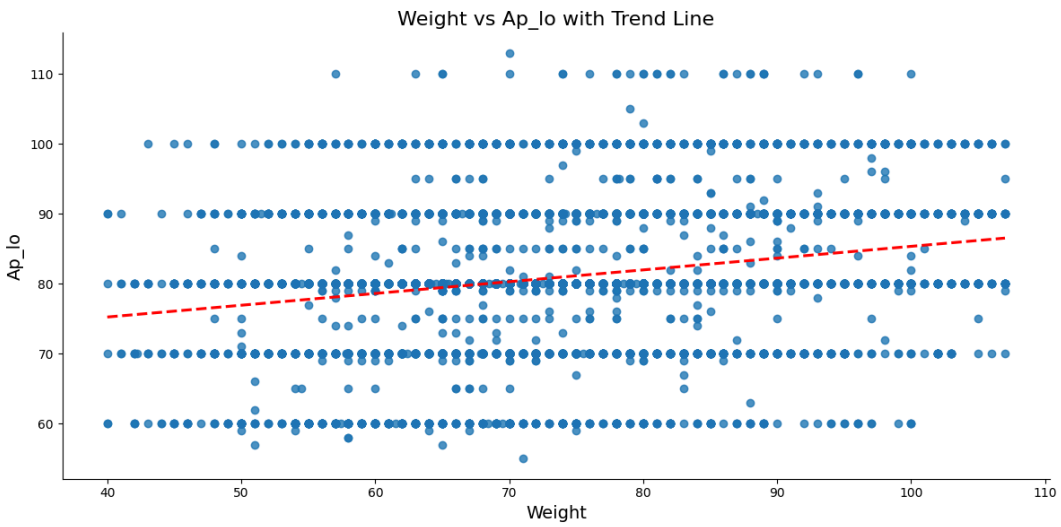

Figure 36: Scatterplot of Weight and Ap_lo relationship

- Weight and Ap_hi: The correlation value is 0.25, meaning that people who weigh more likely to have higher ap_hi.

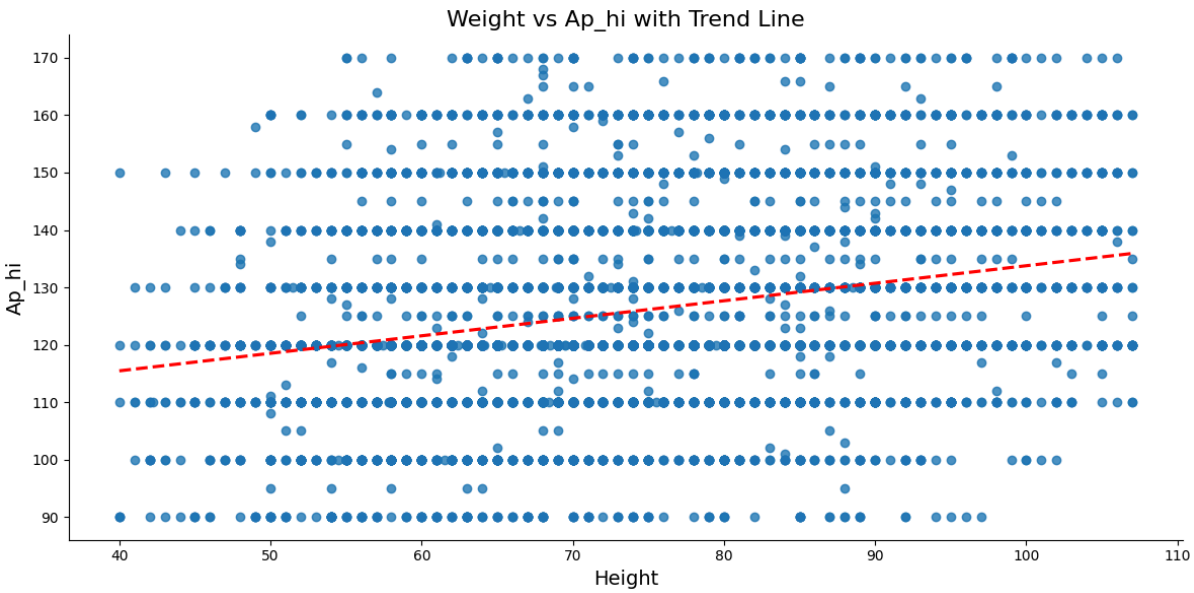

Figure 37: Scatterplot of Weight and Ap_hi relationship

5. Ap_hi-relation correlation
   - Ap_hi and Cardio: The correlation value is 0.44 and is a strong positive correlation, meaning that people who have higher ap_hi will have a higher risk of cardiovascular issues.

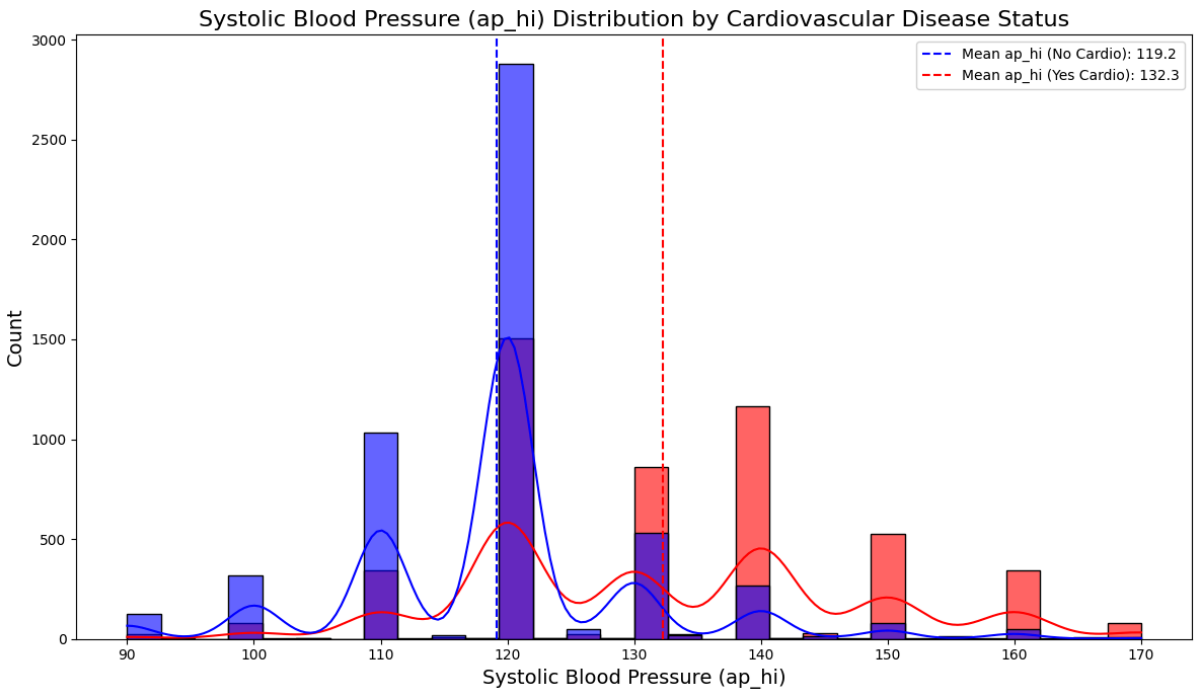

Figure 38: ap_hi distribution by cardiovascular disease

- Ap_hi and Cholesterol: The correlation value is 0.19 and is a positive correlation, meaning that people who have higher ap_hi are likely to have above normal cholesterol levels.

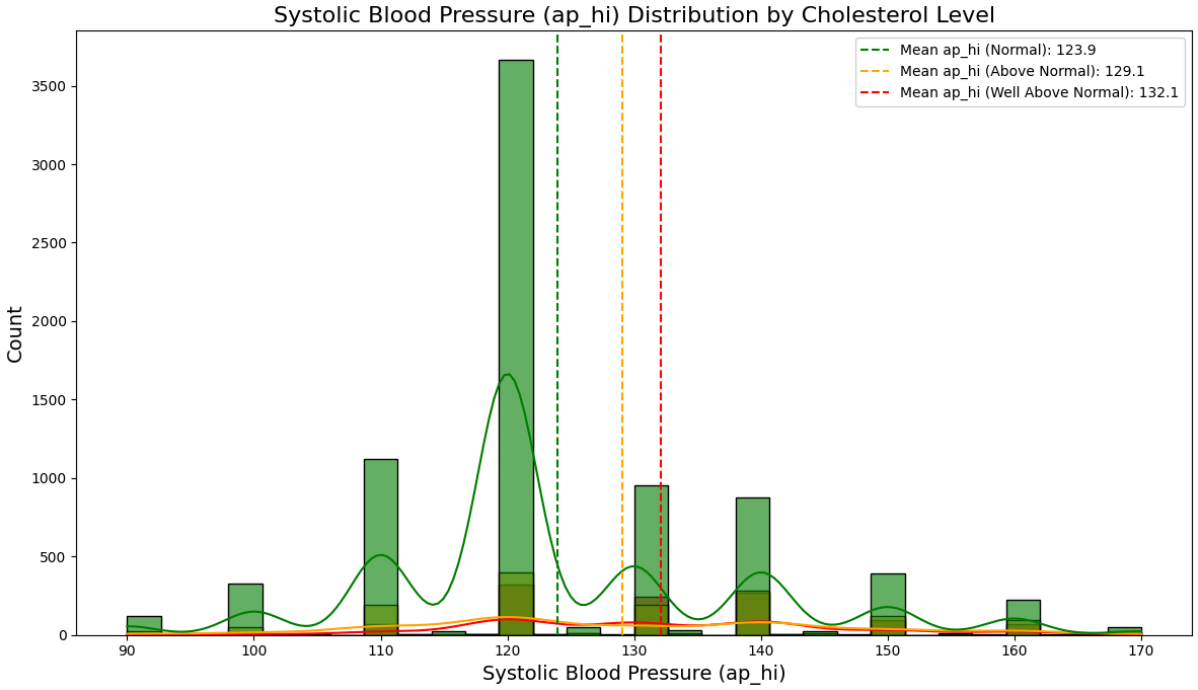

Figure 39: ap_hi distribution by cholesterol level

- Ap_hi and Ap_lo: The correlation value is 0.72 and is a strong positive correlation, meaning that people with high Ap_hi will also have high Ap_lo, which is expected because they should be related.

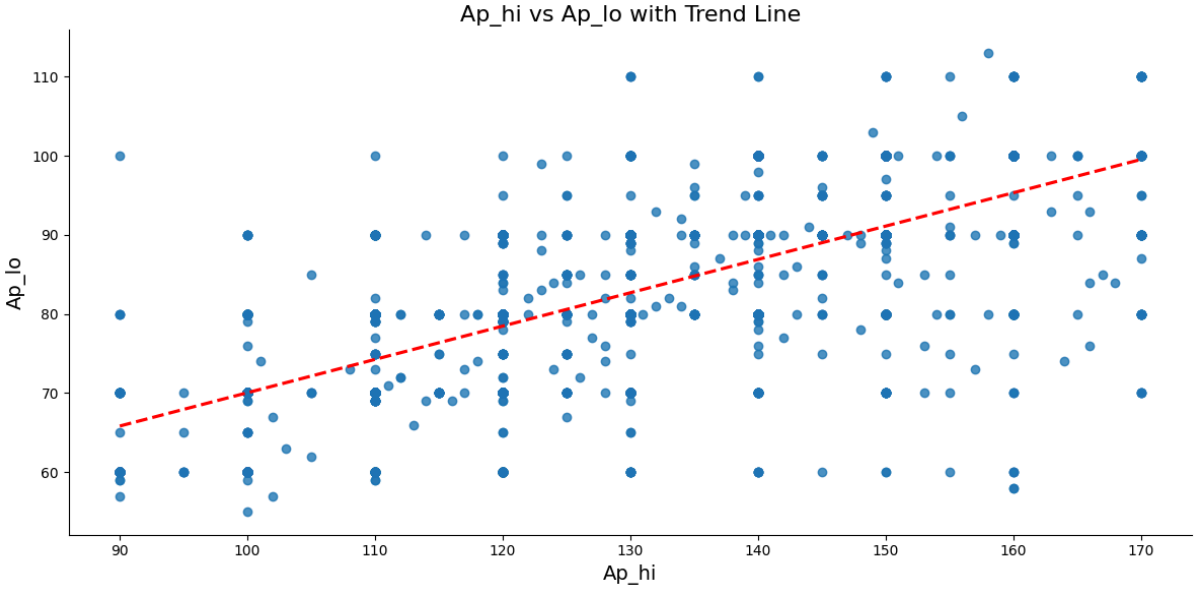

Figure 40: Scatterplot of Ap_hi and Ap_lo relationship

6. Ap_lo-realation correlation
    - Ap_lo and Cardio: The correlation value is 0.34 and is a positive correlation, meaning that people who have higher diastolic blood pressure will have a higher risk of cardiovascular issues.

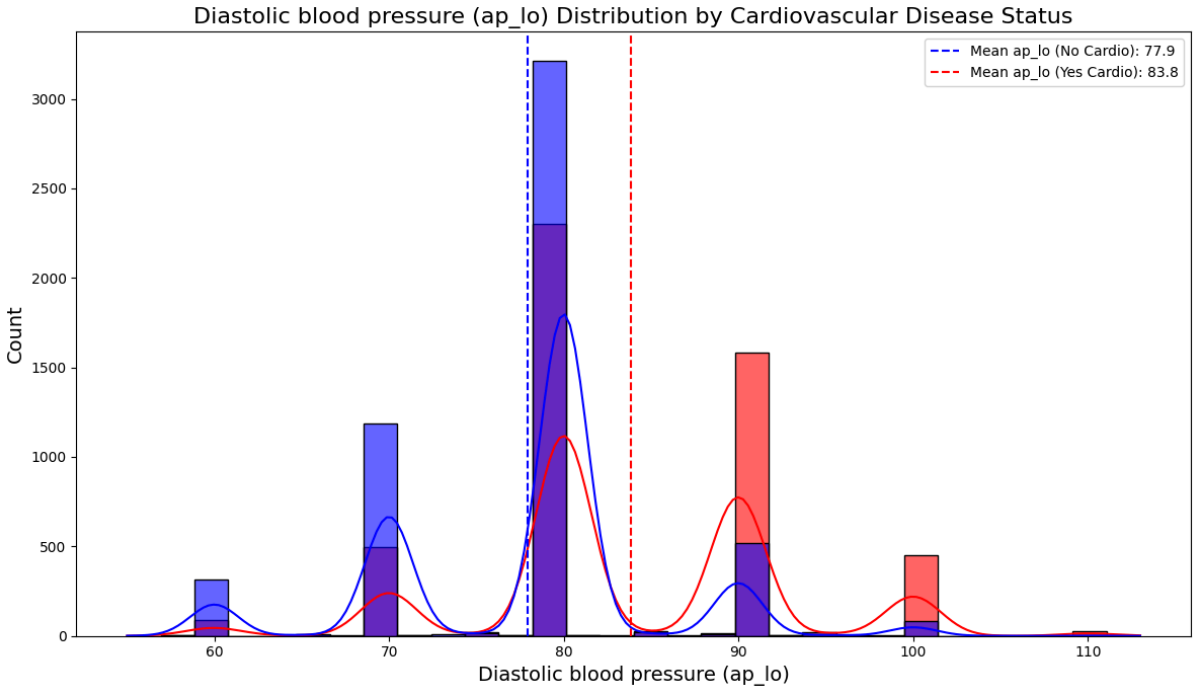

Figure 41: ap_lo distribution by cardiovascular disease

- Ap_lo and Cholesterol: The correlation value is 0.23 and is a positive correlation, meaning that people who have higher diastolic blood pressure tend to have above normal cholesterol levels.

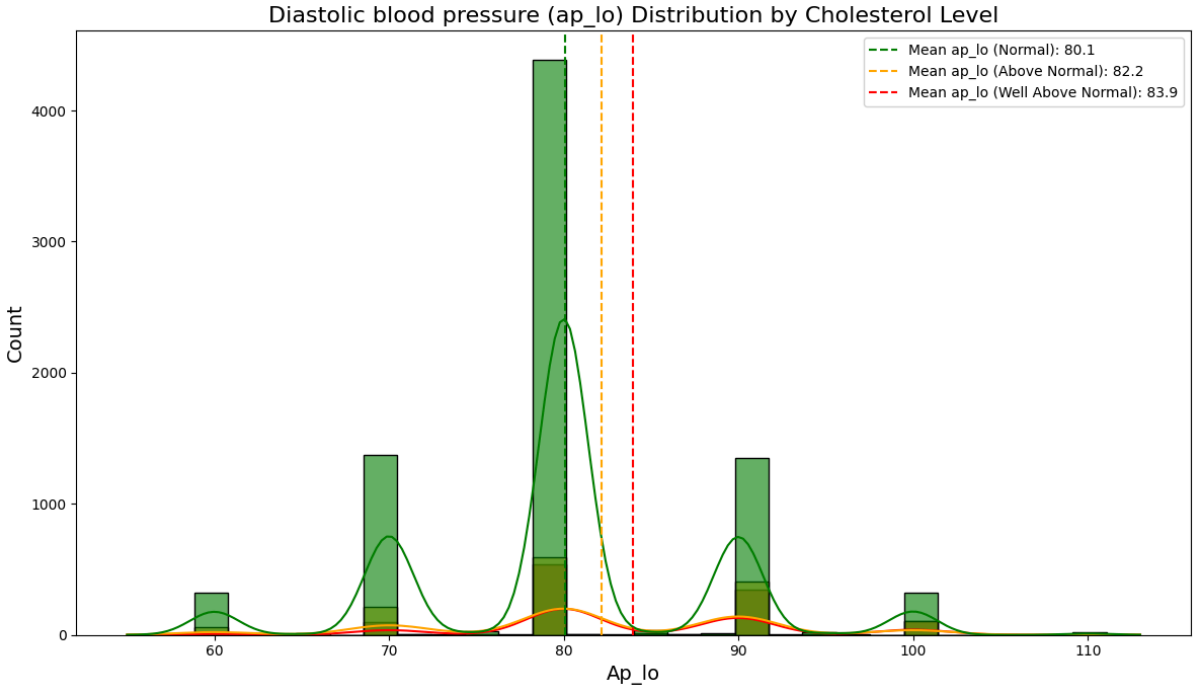

Figure 42: ap_lo distribution by cholesterol level

7. Cholesterol-relation correlation
    - Cholesterol and Cardio: The correlation value is 0.23 and is a positive correlation, meaning that people who have above normal levels of cholesterol may have a higher risk of cardiovascular issues.

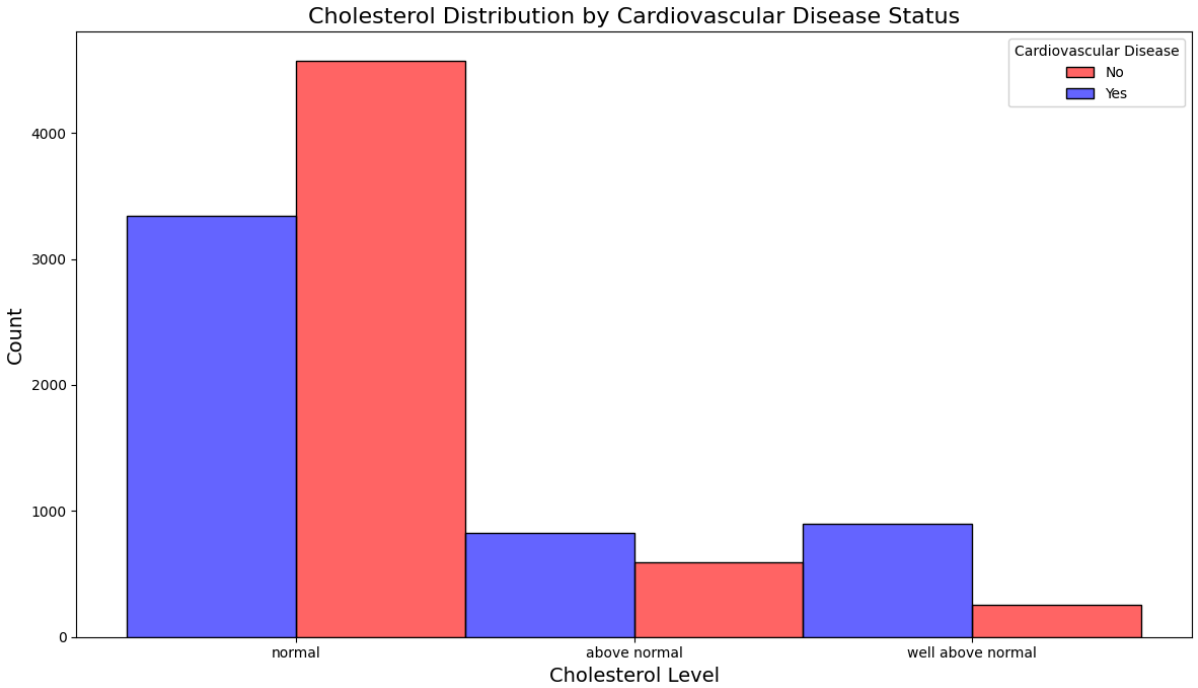

Figure 43: Cholesterol level distribution by cardiovascular disease

- Cholesterol and Gluc: The correlation value is 0.43 and is a strong positive correlation, meaning that most people with above normal levels of cholesterol also have above normal levels of glucose.

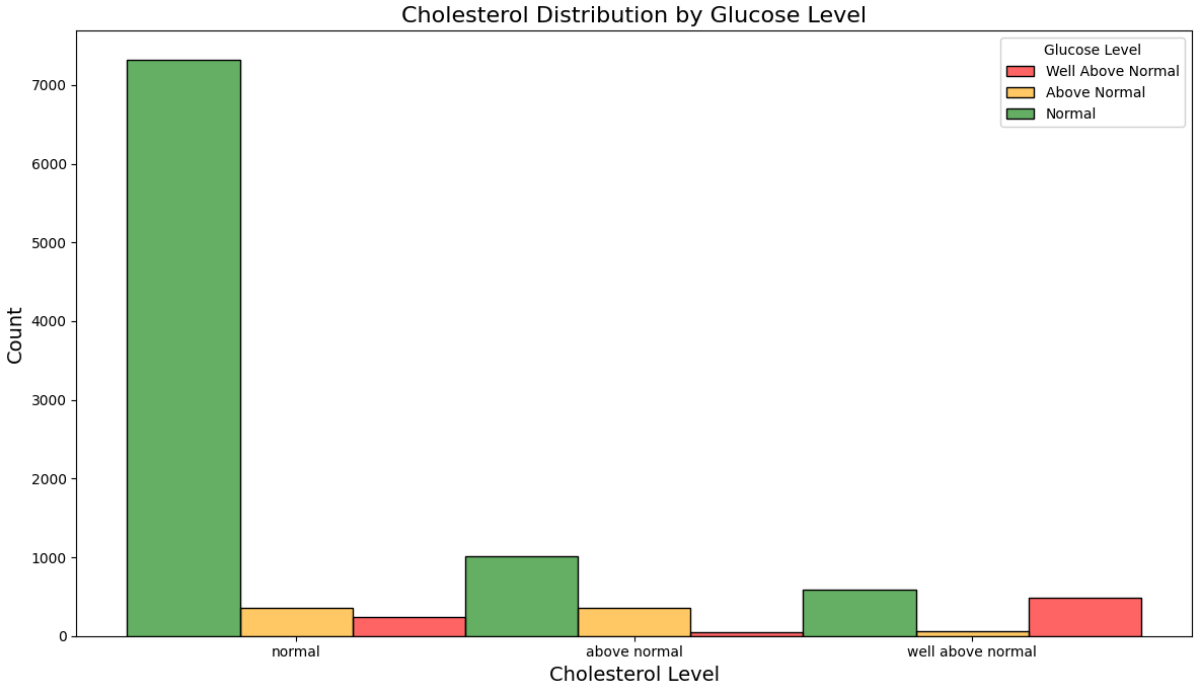

Figure 44: Cholesterol level distribution by Glucose level

8. Other Notable Relationships
    - Gluc and Cardio: The correlation value is 0.09, meaning that people above normal levels of glucose tend to have a higher risk of cardiovascular issues.

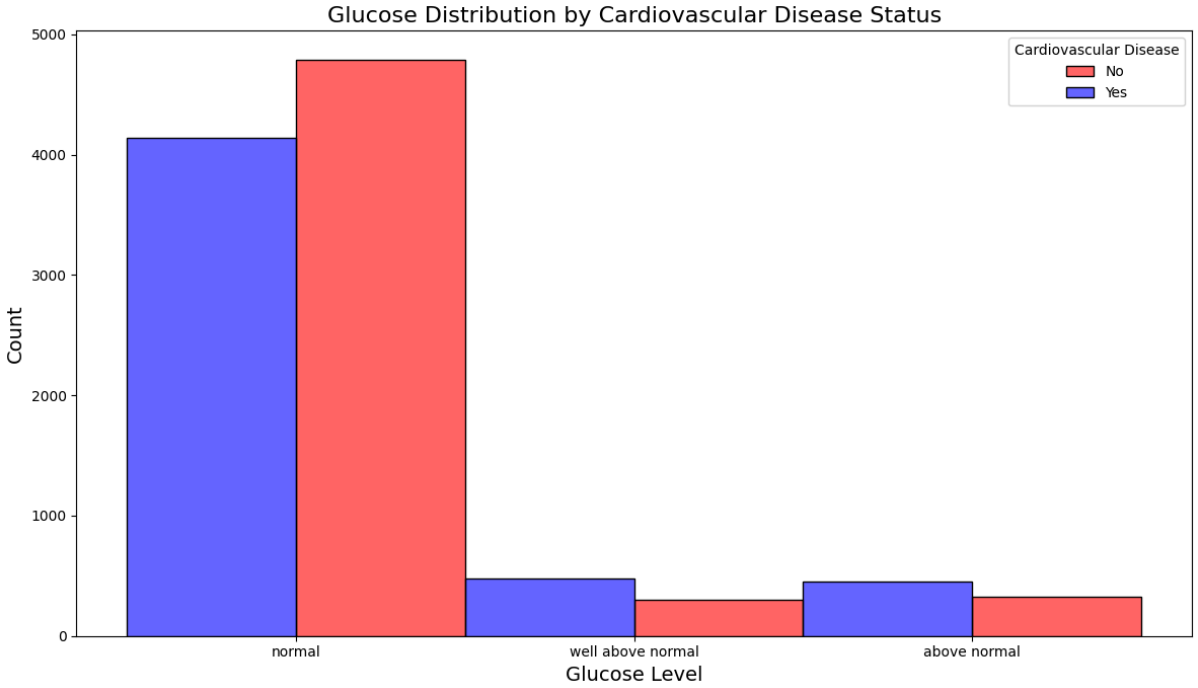

Figure 45: Glucose level distribution by Cardiovascular disease

- Smoke and Alco: The correlation value is 0.33 and is a positive correlation, meaning that people who smoke are more likely to take alcohol than people who don't.

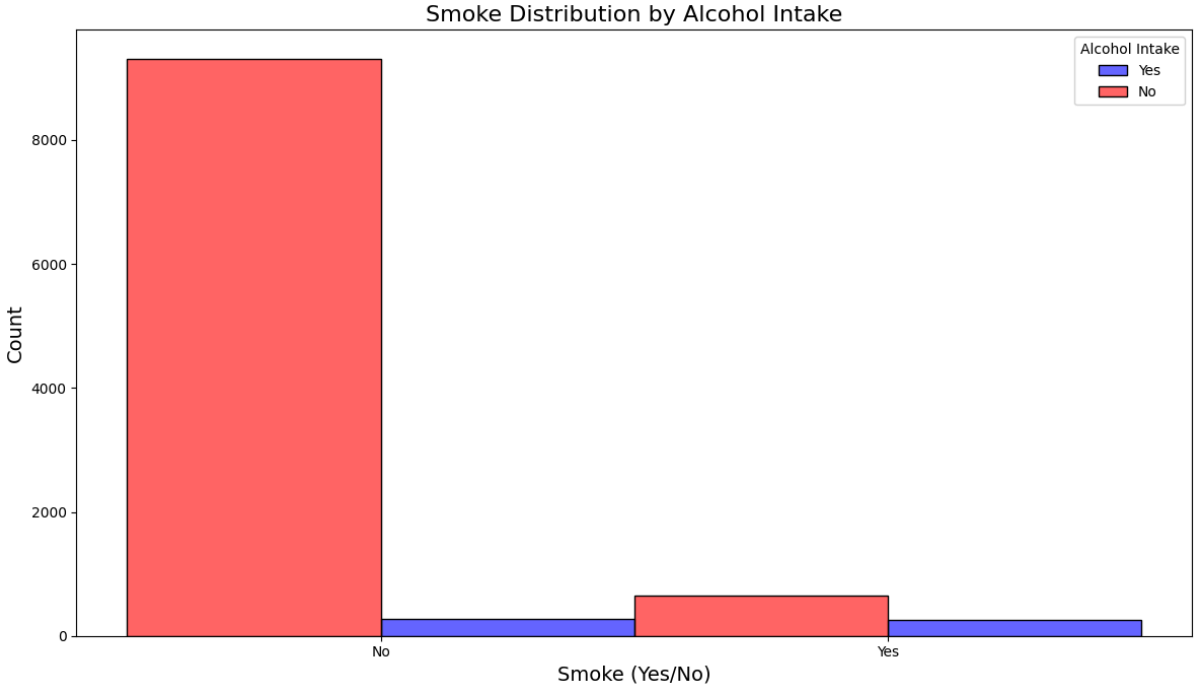

Figure 46: Smoke distribution by alcohol intake

Identify clusters of similar instances.
Cluster Analysis:

1. Cluster 0 (6143 samples)
    - Youngest group (average 18,962 days or 52 years old).
    - Lightest weight (69.59 kg).
    - Lowest systolic and diastolic blood pressure (117.97 / 76.83).
    - Lowest cholesterol (1.10) and glucose levels (1.07).
    - More females.
    - Lowest smoking and alcohol consumption rates.
    - Lowest cardiovascular disease prevalence (33.03%).

2. Cluster 1 (1005 samples)
    - Middle-aged group (19,117 days or 52.4 years old).
    - Higher weight (75.96 kg).
    - Higher blood pressure (126.51 / 81.44).
    - Higher cholesterol (1.36) and glucose levels (1.20).
    - More males.
    - Taller individuals on average (169.46 cm).
    - Highest smoking and alcohol consumption rates.
    - Higher cardiovascular disease prevalence (45.67%).

3. Cluster 2 (3340 samples)
    - Oldest group (20,404 days or 55.9 years old).
    - Heaviest weight (78.15 kg).
    - Highest blood pressure (139.05 / 87.85).
    - Highest cholesterol (1.82) and glucose levels (1.52).
    - More females.
    - Smoking and alcohol consumption are slightly higher than Cluster 0 but lower than Cluster 1.
    - Highest cardiovascular disease prevalence (77.25%).

cluster	age	weight	ap_hi	ap_lo	cholesterol	gluc	gender	height	smoke	alco	active	Cardio (%)
0	18962.33	69.59326	117.9725	76.83347	1.101579	1.06658	1.277389	163.953	0	0.012535	0.202507	33.02946
1	19117.08	75.96358	126.5184	81.44876	1.362189	1.202985	1.898507	169.4587	0.884577	0.371144	0.371144	45.67164
2	20404.75	78.15623	139.0536	87.85868	1.817964	1.516766	1.285629	163.5246	0.002994	0.025749	0.203892	77.24551

Figure 47: Information of each attribute in each cluster

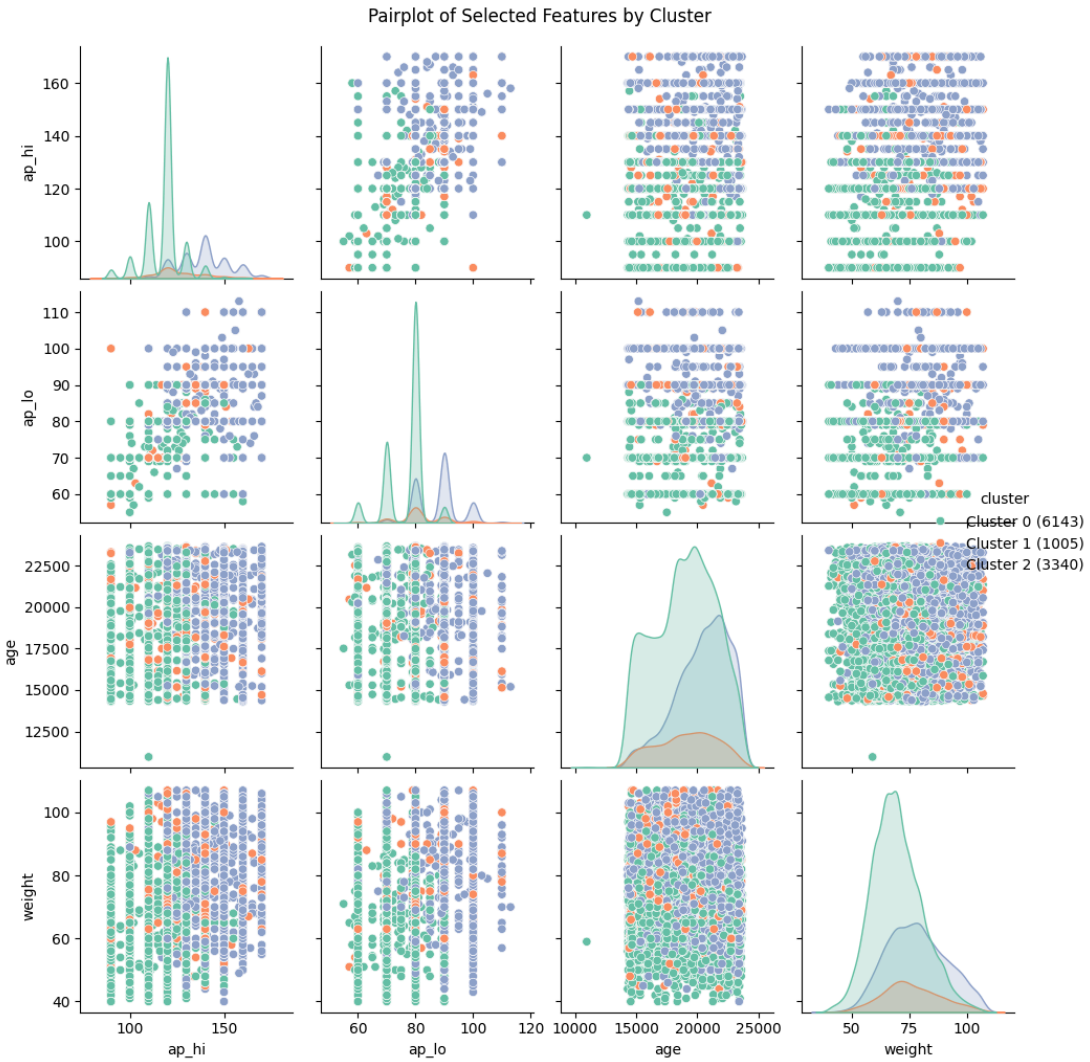

Figure 48: Clusters of the attributes using Pairplot

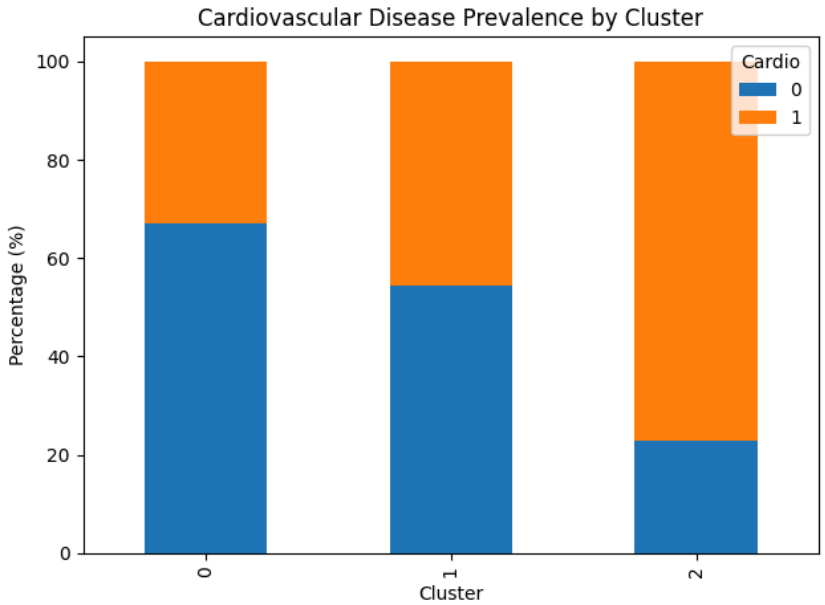

Figure 49: Percentage of patients with cardiovascular disease in each cluster

1B. Data preprocessing

Before doing any technique, I think we have to remove the rows with outliers from the dataset because the outliers can skew the bin range and distort data distribution. For Equi-width binning, the outliers will stretch the bin width, making some bins empty while other bins contain most data points. For Equi-depth binning, outliers may disrupt equal distribution across bins. In addition, without outliers, the binning method will better represent the majority of data points and prevent misleading results when analyzing patterns.

Using the Interquartile Range Method to remove the outliers from column height, weight, ap_hi, and ap_lo. As I have indicated from part 1A, Initial data exploration, the outliers will be outside the bounds of:

Height: [131.5, 197.5]

Weight: [39.5, 107.5]

Ap_hi: [90, 170]

But for the Ap_lo since using the normal formula and got the bounding [65, 105], I think the lower bound is higher than the reality, so I changed the coefficient to 2.5 so I get the bounding [55, 105].

Figure 51 shows the number of outliers below. After determining the bounds of the attributes, I decided to remove the row with the outliers because there are not many rows that contain the outliers, so removing them will not affect the analysis.

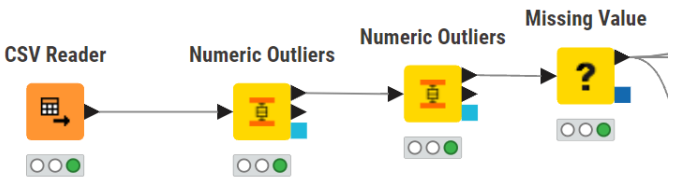

Figure 50: KNIME workflow for removing the outliers

Outli... String	Memb... Number ...	Outlier count Number (integer)
height	11200	19
ap_lo	11200	245

Outl... String	Me... Numb...	Outlier count Number (integer)
weight	11200	295
ap_hi	11200	247

Figure 51: Number of outliers in each attribute

1. Binning the height column using Equi-width binning and Equi-depth binning method.

After removing the rows with outliers by the Interquartile Range Method, to determine the number of bins for Equi-width binning and Equi-depth binning of the height attribute, I use the Rice Rule because this method is specifically useful for larger datasets. The equation is

$$k = 2\times\sqrt[3]{n}$$

After applying the Rice Rule, I got k = 45 bins.

To apply the Equi-width and the Equi-depth binning technique, I use the Auto-Binner node to automatically divide the height attribute into bins. For the Equi-width binning, the node will calculate the range of each bin by

Bin Width = (Max Value - Min Value) / Number of Bins

Then assign the binning to each value.

For the Equi-depth binning, the node will calculate the number of data points in each bin, then automatically assign data points so that each bin contains approximately the same number of observations. For example, if the dataset contains 100 height values and we want 5 bins, each bin should contain about 100/5 = 20 values. The first bin will have the smallest 20 values. The second bin will have the next 20 values. This process continues until all data points are assigned. The bin boundaries are based on the values at these cut-off points.

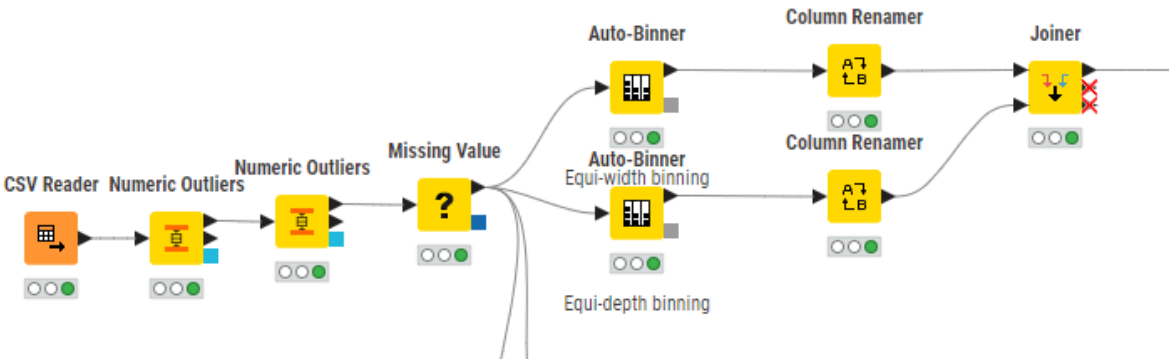

Figure 52: KNIME workflow for binning the height attribute

2. Normalize the weight column using min-max and z-score normalization.

To normalize the weight attribute, Min-max normalization and z-score normalization are used. As shown in Figure 50, I used KNIME to do this task. First, I use the preprocessed table that has been removed the row with outliers, using a column splitter node to split the ID and weight out of the table, then normalize the weight column using a Normalizer node by min-max normalization and z-score normalization. After that, rename each column and join them together. Finally, join the normalization table with the original table and save it in the CSV file.

The differences between min-max normalization and z-score normalization are the formula, range of values, effect on data, and sensitivity to outliers.

Method	formula	range of values	effect on data	sensitivity to outliers
Min-max normalization	$X'=(X-X_{min}) / (X_{max}-X_{min})$	[0,1]	Preserves the original distribution but rescales it	Outliers affect min and max values
Z-score normalization	$X' = (X-\mu)/ \sigma$	Mean-centered. Range is determined by the range of the value.	Centers data around 0, removing the mean and adjusting the variance	less affected due to the mean and standard deviation

By considering these differences, I found that you should use min-max normalization when you need the data in a fixed range, like in neural networks and image processing, and use z-score normalization when data has outliers or follows a normal distribution, like regression in machine learning.

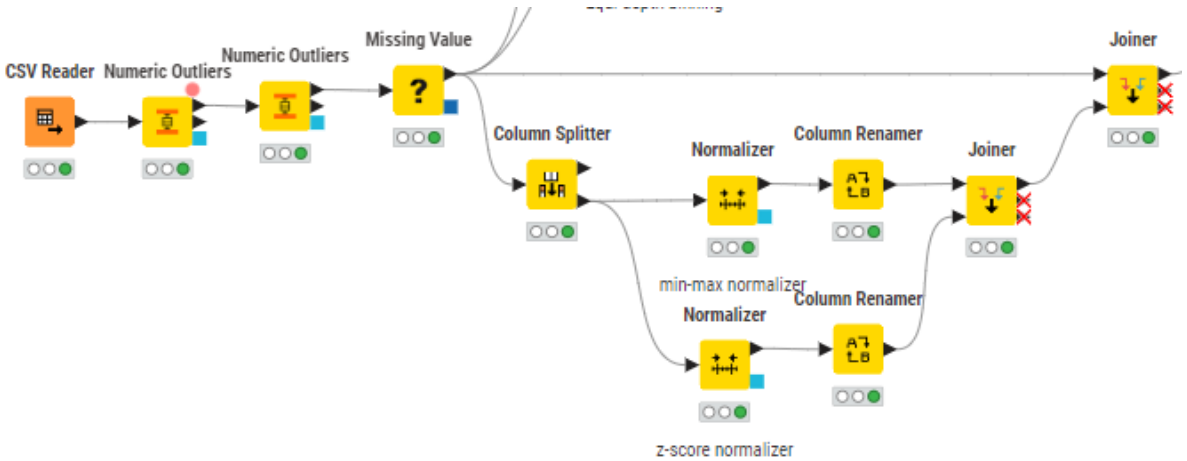

Figure 53: KNIME workflow for normalization

3. Discretise the age attribute into categories.

To discretise the age attribute into categories, I have to transform the age in days into years by using the Expression node($age/365). This node will transform age in days to years. After the transformation, I use Numeric Binner node to categorize the age in year, Young (ages 21 and younger), Early Adulthood (ages 22-34), early Middle Age (ages 35-44), late Middle Age (ages 45-64), and Late Adulthood (ages 65 and older) and set the column name to Age_binned.

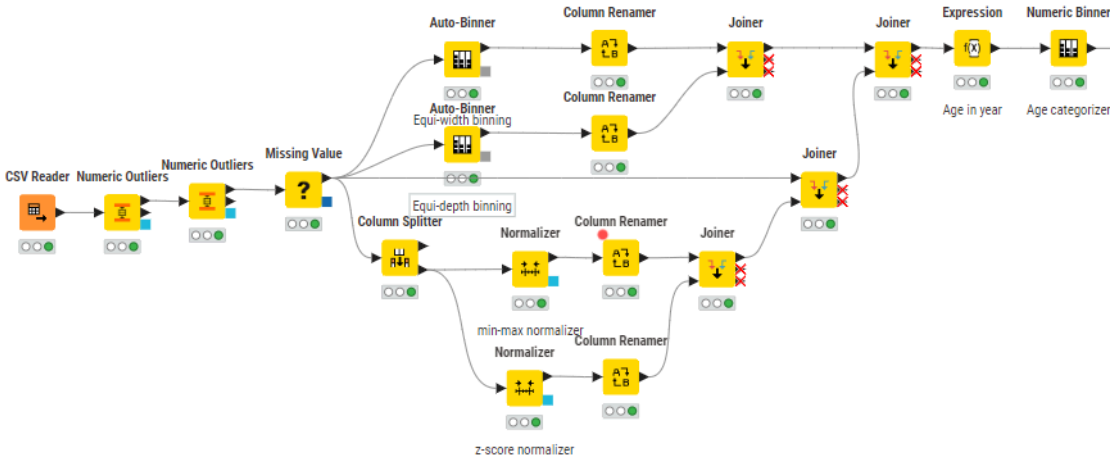

Figure 54: KNIME workflow for discretizing the age

4. Binarize the smoke variable [with values "0" or "1"].

To binarize the smoke variable, I use the Category to Number node in KNIME. This node will convert categorical variables into numerical representations. The smoke variable has categories “Yes” and “No”; the node will map them to 1 and 0.

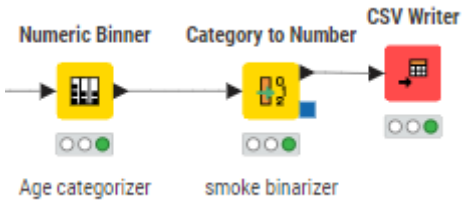

Figure 55: KNIME workflow for binarizing the smoke variable

1C. Summary

From the data analytics, I have found the factors that directly affect cardiovascular disease, from most to least, are: systolic blood pressure(ap_hi), diastolic blood pressure(ap_lo), age, cholesterol levels, weight, and glucose levels, respectively.

Other factors such as gender, height, smoke, alcohol intake(alco), and physical activities(active) were not found to directly affect cardiovascular disease. However, some factors may indirectly affect the development of cardiovascular disease, such as gender and height attributes that have relations with the weight attribute; these attributes (gender and height) also have a relation between them, and the correlation value is 0.51, which means these attributes affect each other.

There are also other factors that directly affect both gender and height, including smoking and alcohol intake. This shows that weight is related to cardiovascular disease, gender and height are related to weight, while smoking and alcohol intake both have relationships to gender and height. Therefore, smoking and alcohol intake may also affect cardiovascular disease, but it is often an indirect effect through weight and other factors that are related to it.